



WESTERN HOGNOSE · HETERODON NASICUS

# The snake that bluffs, digs, and plays dead.

A complete, fact-checked printable manual for the western hognose: the deep, dry setup this burrower needs, feeding by gram weight, what the hood, the hiss and the death act actually mean, a calm and sourced answer to the venom question, and a state-by-state summary of where you can keep one.

---

## INSIDE

- ✓ The enclosure sized by sex, the dry-air rule, and digging on the cool end
- ✓ Prey by gram weight, and the plan for a hatchling that refuses
- ✓ The venom question, answered from the bite research
- ✓ A state-by-state legal summary, plus sitter, owner and vet logs

47 pages

Intermediate

Print or tablet

Edition 1.1 · Oct 2026



## CONTENTS

# What's inside

Every page is written to be used, not just read once.

### GETTING STARTED

|                         |   |
|-------------------------|---|
| How to Use This Package | 3 |
|-------------------------|---|

### SECTION 01 · PROFILE

|                                        |   |
|----------------------------------------|---|
| Quick Profile & Cost Overview          | 4 |
| Which Hognose This Is, Size & Lifespan | 5 |

### SECTION 02 · HOUSING & ENVIRONMENT

|                                             |    |
|---------------------------------------------|----|
| The Enclosure, Sized by Sex & Age           | 6  |
| The Temperature Gradient                    | 7  |
| Heat Sources, Thermostats & Probes          | 8  |
| Dry Air, the Humid Hide & Ultraviolet Light | 9  |
| Deep Bedding, Digging & Cleaning            | 10 |

### SECTION 03 · FEEDING

|                                               |    |
|-----------------------------------------------|----|
| Prey Size & Feeding Schedule                  | 11 |
| Frozen Prey: Thawing, Storage & Where to Feed | 12 |
| The Refusing Hognose                          | 13 |

### SECTION 04 · HANDLING & BEHAVIOR

|                                 |    |
|---------------------------------|----|
| Handling & Hygiene              | 14 |
| Hooding, Hissing & Playing Dead | 15 |
| The Venom Question & Bites      | 16 |
| Enrichment & Common Mistakes    | 17 |

### SECTION 05 · ARRIVAL & LIFE STAGES

|                                                      |    |
|------------------------------------------------------|----|
| Choosing a Hognose, Quarantine & the First Vet Visit | 18 |
| Telling the Sex, Weight & Body Condition             | 19 |
| Females, Eggs & Eggs That Get Stuck                  | 20 |
| The Winter Slowdown                                  | 21 |

### SECTION 06 · THE LAW

|                                    |    |
|------------------------------------|----|
| Is a Hognose Legal Where You Live? | 22 |
| States With Conditions             | 23 |
| States Where It Is Legal           | 24 |

### SECTION 07 · HEALTH & COMMON ISSUES

|                                                        |    |
|--------------------------------------------------------|----|
| Health Red Flags & Finding a Vet                       | 25 |
| Respiratory & Belly-Scale Infections                   | 26 |
| Blockages, Regurgitated Meals & Obesity                | 27 |
| Mites & Internal Parasites                             | 28 |
| Shedding, Mouth Infections, Burns & Tissue at the Vent | 29 |
| Reading Droppings & Hydration                          | 30 |

### SECTION 08 · QUICK REFERENCE

|                                |    |
|--------------------------------|----|
| Setup Checklist & Targets      | 31 |
| Emergency & Quick Targets Card | 32 |

### SECTION 09 · OWNER TOOLS

|                                   |    |
|-----------------------------------|----|
| Budget: Setup & Shopping List     | 33 |
| Budget: Yearly Costs              | 34 |
| First 30 Days                     | 35 |
| Symptom Quick Reference           | 36 |
| Daily, Weekly & Seasonal Routine  | 37 |
| Power Outages, Travel & Transport | 38 |
| Pet-Sitter Sheet                  | 39 |
| Owner Log                         | 40 |
| Equipment, Quarantine & Vet Log   | 41 |

### REFERENCE

|                            |    |
|----------------------------|----|
| Glossary, A to D           | 42 |
| Glossary, E to N           | 43 |
| Glossary, O to Z           | 44 |
| Sources                    | 45 |
| Where the Sources Disagree | 46 |
| Version History & About    | 47 |

## START HERE

# How to use this package

Three ways to get value from it, depending on where you are right now.

### UNITS USED THROUGHOUT

Every temperature and size is given twice. Temperatures appear in degrees Fahrenheit (°F) with degrees Celsius (°C) beside them. Sizes appear in inches (in) and feet (ft), with centimeters (cm) beside them. Snakes and their food are weighed in grams (g), the unit a kitchen scale shows. Humidity is a percentage (%) of how much moisture the air could hold. Prices are in United States dollars.

### If you haven't bought a hognose yet

Read the law section first, starting on page 22. In most states a western hognose raises no legal question at all, but in a handful it is banned, capped or permit-only, and in several more the answer depends on which of the three hognose species you buy. Then read pages 4 through 10 before buying any equipment. Two decisions are expensive to get wrong: the enclosure size on page 6, which depends on whether you buy a male or a female, and the heat source on page 8, which has to run through a thermostat, a controller that cuts the heat at a set temperature. The Budget and Shopping List on page 33 then lets you buy once instead of twice.

### If your hognose is already home

Start by auditing what you have against the Setup Checklist in the print list below. Then read page 9: this species wants dry air, and an enclosure kept damp is the root of the two most common illnesses in it. If your snake hoods, hisses or rolls over to play dead, page 15 explains why that is normal. If it is refusing food, page 13 comes before any worry. If anything already looks wrong, go straight to Health Red Flags on page 25.

### Print these first

- |                                                                    |                                                                   |
|--------------------------------------------------------------------|-------------------------------------------------------------------|
| <input type="checkbox"/> Page 31: Setup Checklist & Targets        | <input type="checkbox"/> Page 39: Pet-Sitter Sheet                |
| <input type="checkbox"/> Page 32: Emergency & Quick Targets Card   | <input type="checkbox"/> Page 40: Owner Log                       |
| <input type="checkbox"/> Page 36: Symptom Quick Reference          | <input type="checkbox"/> Page 41: Equipment, Quarantine & Vet Log |
| <input type="checkbox"/> Page 37: Daily, Weekly & Seasonal Routine | <input type="checkbox"/> Page 11: Prey Size & Feeding Schedule    |

### WHY THIS PACKAGE EXISTS

The western hognose has a reputation built on two things: a theatrical defensive act, and the word "venomous." The act makes owners think a calm snake is an aggressive one, and the word makes owners think a mildly venomous snake is a dangerous one. Neither is true, and both are easier to live with once you know what is going on. The care itself is simple once three things are right: dry air, deep digging substrate (the loose material on the enclosure floor), and feeding an adult less often than a growing juvenile. Every figure here comes from veterinary and scientific sources, listed on page 45, and every temperature and dimension is given in both imperial and metric units.

### NOT A SUBSTITUTE FOR

A veterinarian who sees reptiles. This is a husbandry reference for day-to-day care, and anything urgent is a professional call, not a page in this book. Find a reptile vet before you need one, and write the number on the emergency card. Nothing in the law section is legal advice either.

## Quick profile

Everything at a glance before you read the detail.

|                      |                                                                                                                                                                                                                         |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Species covered      | The western hognose, <i>Heterodon nasicus</i> , also called the plains hognose. It is the hognose the pet trade sells. The eastern and southern hognoses are different species, and the difference matters for the law. |
| Native range         | The North American plains, from southern Canada to northern Mexico: shortgrass prairie, dry rocky land, and sandy soil it can dig in                                                                                    |
| Adult size           | Females run much larger than males. A typical adult male is 16 to 24 in (41 to 61 cm); a typical adult female 21 to 30 in (53 to 76 cm).                                                                                |
| Captive lifespan     | 10 to 15 years is the usual range, with 15 to 20 reported in captivity                                                                                                                                                  |
| Activity             | Active by day, most of all in the morning and late afternoon, and a dedicated burrower                                                                                                                                  |
| Diet                 | Toads in the wild. In captivity, frozen-thawed mice, fed every 5 to 7 days while young and about every two weeks as an adult. Page 11                                                                                   |
| Enclosure, one adult | Male: 20-gallon equivalent, about 30×13×13 in (76×33×33 cm). Female: 40-gallon equivalent, 36×18×16 in (91×46×41 cm). Floor space over height. Page 6                                                                   |
| Temperature          | Basking 90 to 95°F (32 to 35°C), cool side 70 to 75°F (21 to 24°C), nights at room temperature, no colder than 60°F (16°C). Page 7                                                                                      |
| Humidity             | 30 to 50%: dry. One humid hide (a box or cave the snake can get inside) for shedding, and no misting the enclosure. Page 9                                                                                              |
| Lighting             | A 12-hour day. UVB (ultraviolet B, the part of sunlight a reptile's skin uses to make vitamin D3) is not required, but a low-output tube suits this day-active snake.                                                   |
| Substrate            | 3 to 6 in (8 to 15 cm) of loose, dry, diggable substrate, about 70% soil-based to 30% reptile-safe sand. Page 10                                                                                                        |
| Venom                | Rear-fanged and mildly venomous. Not considered dangerous to a healthy person. Page 16                                                                                                                                  |
| Conservation status  | Least Concern, the lowest-risk category on the global Red List of threatened species, but listed as threatened in some states, Illinois among them. Buy captive-bred                                                    |

## Cost overview

### Initial setup

The snake is \$50 to \$100 for a normal, wild-type animal (natural color and pattern), and far more for a rare color form. Females typically cost \$50 to \$200 more than males. Itemized, the equipment comes to **\$390 to \$850**, and the enclosure is the biggest single line. With the snake, a first exam and a fecal test (a stool check for parasites), plan on **\$505 to \$1,080**. Page 33 has every line.

### Ongoing

Frozen mice, substrate, electricity, a UVB tube if you run one, and a yearly wellness exam come to **\$200 to \$365 a year**, about \$15 to \$35 a month. A respiratory infection or an impaction can run into the hundreds to treat, and an X-ray image, called a radiograph, is \$100 to \$250 a view.

### THE NUMBER THAT SHOULD DRIVE THE DECISION

Not the price. The lifespan. A hognose bought for a ten-year-old can still be in the house when that child leaves for college, and some live to 20. The purchase is the cheap part. The routine on page 37, and a reptile vet for every one of those years, is what you are signing up for.

# Which hognose this is, size & lifespan

"Hognose" is a common name shared by several species. The law and the care both depend on which one you have. Three hognose species live in the United States, and a fourth, the Mexican hognose, reaches into the southwest. The one sold as a pet is almost always the **western hognose, *Heterodon nasicus***, which some state laws call the plains hog-nosed snake. Its smaller size, milder bluffing and flexible diet made it the hognose most keepers settled on, and most animals sold now are captive-bred rather than collected. This package is written for the western hognose only.

| SPECIES                                                     | WHERE IT FITS                                                                                                                     | WHY IT MATTERS TO YOU                                                                                                    |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Western (plains) hognose</b><br><i>Heterodon nasicus</i> | The pet-trade species. Dry prairie and sandy ground                                                                               | Every figure in this package. Several state laws name it directly, some to allow it and some to restrict it              |
| <b>Eastern hognose</b><br><i>Heterodon platirhinos</i>      | Native across much of the eastern half of the country. Its care differs: more humidity and a cooler basking spot than the western | Protected or restricted as a native species in many states that leave the western alone. Never collect one from the wild |
| <b>Southern hognose</b><br><i>Heterodon simus</i>           | A southeastern native                                                                                                             | Protected by name in Alabama, and on Georgia's list of natives that may not be kept as pets                              |

## GET THE SPECIES IN WRITING

State rules are written to the scientific name, and the common names overlap: "plains hognose" and "western hognose" are the same snake. Ask for a receipt that names *Heterodon nasicus* and says captive-bred. In several states that one line is the difference between a legal pet and a prohibited one. Pages 22 to 24.

## How big, and how heavy

| STAGE        | LENGTH                                                                                        | WEIGHT                                                          |
|--------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Hatchling    | 5.5 to 7 in (14 to 18 cm)                                                                     | Usually 4 to 7 g. Under 3 g happens, and 13 to 14 g is a record |
| Adult male   | Typically 16 to 24 in (41 to 61 cm), many around 20 in (51 cm). Over 25 in (64 cm) is unusual | Small. Adult males are commonly fed as 40 to 70 g snakes        |
| Adult female | Typically 21 to 30 in (53 to 76 cm)                                                           | 150 to 200 g in a young adult, 220 to 400 g in an older one     |

Published ranges vary. Another range is 14 to 37 in (36 to 93 cm), averaging about 20 in (50 cm), at 80 to 350 g, and across the hognose species as a group the range runs 1.5 to 3.5 ft (45 to 107 cm). The weights in the table are a guide, not a target: your own snake's weight trend matters more.

## How long it lives

**10 to 15 years** is the usual captive range, and **15 to 20** is reported in captivity. In the wild, a western hognose lives about 14 years on average, with a reported range of 9 to 19.

## The animal behind the care

The upturned snout is a digging tool. The enlarged scale at its tip, the rostral scale, is shovel-shaped, and the snake uses it to burrow through loose soil after buried toads, its main wild prey. Toads can make up nearly 80% of the diet in some regions. In the wild it spends roughly September to March below ground. Every part of the captive setup follows from that life: dry air, deep loose substrate, and a snake that may slow down in winter.

## SECTION 02 · HOUSING & ENVIRONMENT

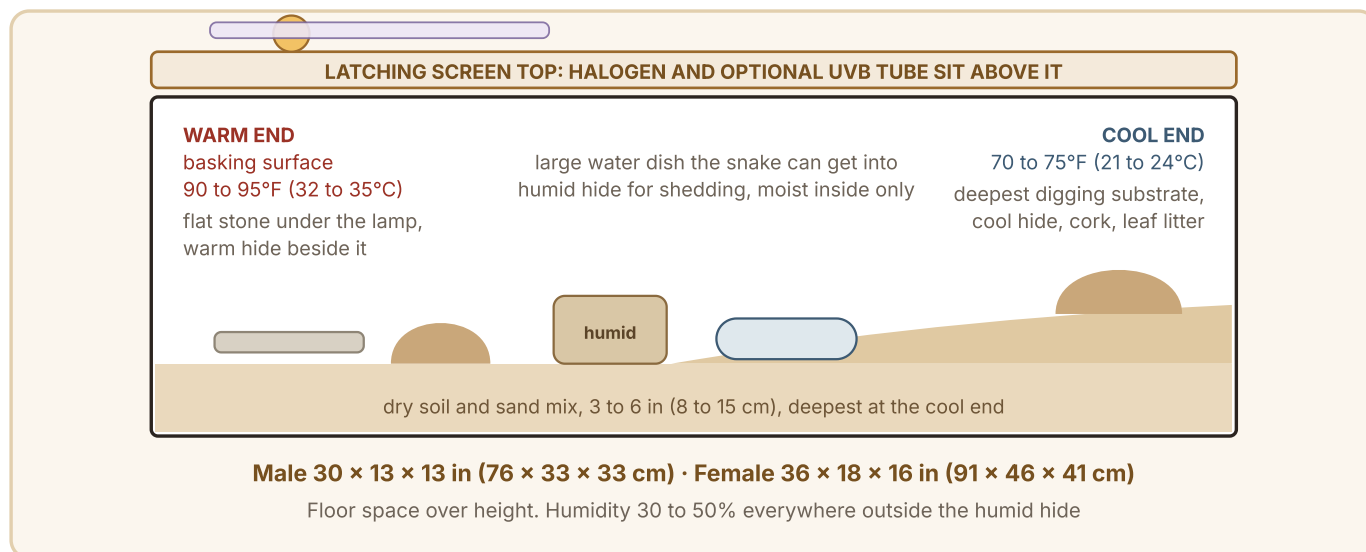
# The enclosure, sized by sex & age

A ground-dwelling burrower measures its home in floor space. Height does almost nothing for it.

The working rule is **about one square foot of floor space per foot of snake length**. Because females grow so much larger, the minimum depends on sex:

| SNAKE                           | MINIMUM ENCLOSURE                                                                                                                               |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Adult male                      | <b>20-gallon equivalent</b> , about 30×13×13 in (76×33×33 cm)                                                                                   |
| Adult female                    | <b>40-gallon equivalent</b> , 36×18×16 in (91×46×41 cm)                                                                                         |
| Either sex, a solid adult setup | A front-opening enclosure of about 2×2×2 ft (61×61×61 cm), or 36×18×18 in (91×46×46 cm)                                                         |
| Hatchling under 6 in (15 cm)    | A 5-gallon enclosure, about 16×8×10 in (41×20×25 cm)                                                                                            |
| Juvenile                        | A 10-gallon enclosure until it is about a year old. Starting in the adult enclosure also works, if it has enough hides and cover to feel secure |

Front-opening PVC (polyvinyl chloride, a rigid plastic) or glass both work. If you use glass, cover three sides to cut stress. The lid has to latch: a hognose pushes at any gap it finds, and on a baby's enclosure the lid needs clamps or latches. **Never use tape**, which is notorious for injuring snakes. Keep one snake per enclosure: sharing an enclosure is one of the listed causes of regurgitation, a meal brought back up.



## Where the enclosure goes

| PLACE                         | WHY                                                                                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Out of the kitchen            | Reptiles carry <i>Salmonella</i> bacteria with no sign of illness, and keeping the snake and its gear away from food is the first hygiene rule. Page 14 |
| Out of a sunny window         | The thermostat controls the heat source, not the sun through the glass. Every degree should come from a controlled source, page 8                       |
| In a room you can keep steady | The same heat source behaves very differently in a cold room and a warm one, so a stable room makes the gradient easy to hold                           |

## The temperature gradient

A hot spot at one end, a cool burrow at the other, and the snake chooses.

| ZONE                                                   | °F               | °C               |
|--------------------------------------------------------|------------------|------------------|
| Basking surface, warm end                              | 90 to 95         | 32 to 35         |
| Middle of the enclosure                                | 75 to 85         | 24 to 29         |
| Cool side, where the deep substrate is                 | 70 to 75         | 21 to 24         |
| Night                                                  | Room temperature | Room temperature |
| <b>Add heat if the night falls below</b>               | <b>60</b>        | <b>16</b>        |
| <b>A fault if the basking surface reads well above</b> | <b>95</b>        | <b>35</b>        |

At night the heat goes off and the enclosure can fall to room temperature, as long as it stays at 60°F (16°C) or above.

### Why a gradient

A snake cannot make its own body heat. It sets its temperature by moving between warm and cool spots, so the enclosure exists to give it a choice. The hot spot is for digesting; the cool end is where this species digs, and where it goes to cool down. Where a reptile sits in its range has real consequences for how it fights infection, which is why a sick snake is kept toward the upper end of its normal range and why an enclosure should not run at the bottom of the range by default.

### Reading it properly

- **Two digital probe thermometers**, one probe on the basking surface and one on the cool end. One reading tells you nothing about a gradient.
- **An infrared thermometer**, the handheld point-and-read kind, to spot-check the surface itself: the basking stone, the substrate, the wall near the heat. It finds a hot spot the fixed probes are not sitting on.
- Check both before the snake goes in, and again whenever anything in the setup changes.

### Night heat is a different setup

The basking light goes off at night, and a hognose needs no night bulb. A normal room is fine overnight as long as the enclosure stays at 60°F (16°C) or above. Only if your room drops below that, add a heat source that gives off no light, such as a heat mat or a ceramic heat emitter (a screw-in heater with no visible glow), on its own thermostat.

### Signs the gradient is wrong

| YOU NOTICE                                                              | CHECK                                                                                                                      |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Wheezing, open-mouth breathing, mucus                                   | Too cool, too damp, or poor airflow are the causes of respiratory infection. Read every number, then call the vet. Page 26 |
| A meal brought back up                                                  | An enclosure running too cool is one of the listed causes. Page 27                                                         |
| Basking surface well above 95°F (35°C)                                  | A fault, not a feature. Check the probe has not slipped and the thermostat works.                                          |
| Two thermometers on the same spot disagree by more than a degree or two | One has drifted. Replace it rather than averaging them                                                                     |

## Heat sources, thermostats & probes

Whatever heats the enclosure, a thermostat controls it. That one device is what prevents burns.

The usual heat source for this species is a **low-wattage halogen basking bulb** (a bright, focused light bulb that heats the spot under it), mounted over the screen at the warm end and run on a dimming thermostat. A 50-watt halogen spot is a typical choice. Some keepers use an under-tank heat mat instead; the halogen is the better choice. Either way, the thermostat is not optional. Put a flat, light-colored stone or piece of wood under the lamp: wild hognoses bask on rocks and logs, and a flat surface warms the snake more efficiently.

### Match the thermostat to the heat source

| THERMOSTAT                | HOW IT CONTROLS HEAT                                                        | BEST MATCHED TO                                               |
|---------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>On/off</b>             | Cuts power fully at the set temperature, restores it when the reading drops | Heat mats, which change temperature slowly                    |
| <b>Pulse proportional</b> | Sends short pulses of power for a steadier average                          | Ceramic heat emitters, which can overshoot on an on/off cycle |
| <b>Dimming</b>            | Turns the power up and down like a light dimmer                             | Halogen basking bulbs, with no visible flicker                |

These pairings are the usual best match, not a hard rule. What matters most is that the heat source has a controller at all.

### Where the probe goes

A thermostat only regulates the spot its probe measures. **For a heat mat**, the probe goes on the floor of the warm hide, the surface the snake lies on, held with a probe clip or suction cup, never tape: foil tape over the probe throws off its reading, and a snake can stick to exposed adhesive and tear scales. **For an overhead halogen**, makers place the probe differently: on the basking surface under the lamp, or in the air a few inches below the bulb. Follow your own thermostat's instructions, keep the probe at the snake's level rather than up near the fixture, and confirm the result with the infrared thermometer.

#### NEVER WITHOUT A THERMOSTAT, NEVER WITHIN REACH

An unregulated mat or bulb keeps drawing full power with no ceiling, and a snake pressed against an overheated surface does not always move away in time. That is how a burn happens. Keep any overhead source at least **4 to 6 in (10 to 15 cm)** above the highest spot the snake can reach, behind the screen. Skip heat rocks entirely: they burn snakes and heat poorly.

### Plan for the thermostat failing

- **The probe slips.** The most common failure, and a silent one: the display stays accurate for the wrong spot. Look at the probe at every water change.
- **The thermostat sticks on.** Rarer and more dangerous. A second thermometer, away from the thermostat's probe and read daily, is what catches it.
- **Wattage is a starting guess.** The number on the box assumes an average room. Set it up, then measure for a day before the snake goes in.

## Dry air, the humid hide & ultraviolet light

The number that prevents the most illness in this species is a low one.

Keep ambient humidity at **30 to 50%**, dry by the standard of most pet snakes. **Do not mist the enclosure** to raise it. Running the whole enclosure damp is a direct path to respiratory infection and scale rot, two illnesses that both trace back to the same cause. A western hognose comes from prairie and sandy ground, and that makes its setup drier than a ball python's or a corn snake's.

### The one humid spot

Give the snake one **humid hide** for shedding: a hide whose substrate is mixed with a little water, or packed with moist sphagnum moss, while the rest of the enclosure stays dry. Add moisture there only as needed, mostly when a shed is coming (page 29). Set it between the warm and cool ends so the snake can use it at a comfortable temperature.

### Measure it, don't guess it

A digital hygrometer (a humidity meter) is core equipment here, not an extra, because this species' whole setup depends on humidity being confirmed low rather than guessed at. Read it at snake level, away from the water dish and the humid hide.

### A day and a night

Unlike most pet snakes, a hognose is active by day, so it gets a normal day and night: lights on a timer, **12 hours on and 12 off**. An optional seasonal schedule runs longer days in summer and shorter ones in winter.

### Ultraviolet light

UVB is not strictly required, and a hognose will not die without it. But a **low-output UVB tube** suits this day-active species. The specification that fits:

| WHAT                  | SPECIFICATION                                                                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tube                  | A T5 HO tube (T5 is the slim fluorescent tube, 5/8 in across; HO means high output) sold as "5.0" or "6%" strength                                                                                                                 |
| Strength at the snake | UVI (ultraviolet index, a measure of UVB strength at one spot) peaking at 1.1 to 3.0 where the snake basks. Hognoses sit in Ferguson zone 2, the band for partial-sun and occasional baskers, about 0.7 to 1.0 across the gradient |
| Length                | About two-thirds of the enclosure length: about 24 in (61 cm) for a 36 in (91 cm) enclosure, at the warm end                                                                                                                       |
| Distance              | About 10 to 14 in (25 to 35 cm) from the tube to the snake's back on the basking surface, with the tube over the screen. Ordinary glass and most plastics block most UVB, so the tube never sits above a glass lid                 |
| Timer and replacement | On the same 12-hour timer as the light. Replace the tube on the maker's schedule, 6 to 12 months depending on the model and no longer than 12, even if it still lights up: UVB fades before the visible light does                 |
| Meter                 | A UV index meter is the only way to know the tube is at the right height and when to replace it; UV cards show total UV, not the UVB that makes vitamin D3.                                                                        |

#### DEEP SUBSTRATE MAKES UVB SAFE

A burrowing snake controls its own exposure: it basks when it wants and goes under the cool-end substrate when it has had enough. That is why the tube goes at the warm end and the digging goes at the cool end.

## Deep bedding, digging & cleaning

For a burrower the substrate is not decoration. It is the part of the enclosure the snake lives in.

Provide **3 to 6 in (8 to 15 cm) of loose, dry, diggable substrate**. A soil and sand mix works well: roughly **70% soil-based substrate to 30% reptile-safe sand** is a common ratio, and aspen or coconut fiber works as the soil-based part. Depth is the variable that matters. Give the snake enough to disappear into completely, not just to push around, and choose a mix that holds a tunnel rather than collapsing. A depth a hognose can vanish into takes more than one small bag. Keep it dry: damp substrate plus this species' preference for arid conditions is a fast route to skin and respiratory problems.

### PUT THE DIGGING ON THE COOL END

In the preference study on page 17, western hognoses chose the enriched side more strongly when it was also the cooler side. So: deepest substrate at the cool end, basking surface at the warm end, cover at both. The snake never has to choose between the temperature it wants and the behavior it wants.

## Furnishings

| ITEM                             | WHY                                                                                                               |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Warm hide and cool hide          | Two secure hides minimum, one at each end, so hiding never costs the snake its preferred temperature              |
| Humid hide                       | For shedding.                                                                                                     |
| Cork bark, branches, leaf litter | Break up the floor so there is cover everywhere, not only in two boxes                                            |
| A large water dish               | Big enough for the snake to get into. It supports shedding and drinking and gives the snake an option it controls |
| Flat basking stone               | Under the lamp at the warm end. Page 8                                                                            |

**Never feed on the substrate.** Feed in a separate container, page 12: a hognose that takes a mouse with a mouthful of substrate attached is the impaction case on page 27.

## Cleaning

| WHEN                 | WHAT                                                                                                                                                                                                                                                 |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| As you see it        | Pull out droppings and urates (the white, chalky part of a reptile's waste)                                                                                                                                                                          |
| Daily                | Empty, wash and refill the water dish with fresh water                                                                                                                                                                                               |
| About every 3 months | Replace the whole substrate even if it looks clean, because urine soaks into it, and disinfect every surface and hide while the enclosure is empty. This longer interval holds only for a bed kept dry with every dropping pulled the day it appears |
| Monthly instead      | For a bed that has gone damp, smells, or hid a dropping for days                                                                                                                                                                                     |

A bioactive setup, a planted substrate with a cleanup crew of tiny invertebrates, is the exception and can run for years without a full change. To disinfect: scrub first, then household bleach at 1:32, which is half a cup per gallon of water, kept wet on the surface for 10 minutes, rinsed until no bleach smell remains, and aired for a full day before the snake goes back. Do it in a utility sink, a dedicated tub or outdoors, never the kitchen sink or a bathtub people use.

## Prey size & feeding schedule

A hatchling can weigh 3 g. At that size the question is not which mouse, but which part of one.

Mice are the staple and the best diet for this species, fed **frozen-thawed**: frozen, then thawed before feeding. Never live: a live rodent can injure a snake that does not subdue it at once, and a 3 g hatchling has no margin for that. Larger individuals sometimes take small rats. Gerbils, hamsters, multimammate mice and chicks are fallbacks for a snake that will take nothing else.

### Prey size for hatchlings, by the snake's weight

The chart runs by weight, which is more useful than age for an animal that arrives at wildly different sizes. Weigh the snake on a kitchen scale in grams.

| SNAKE WEIGHTS    | PREY                                                                                                                                   |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 3 g and under    | A small pinky head. A pinky is a newborn, hairless mouse                                                                               |
| 4 to 6 g         | Extra small pinkies                                                                                                                    |
| 7 to 9 g         | Small to medium pinkies                                                                                                                |
| Larger juveniles | Large pinkies to peach fuzzies, then peach fuzzies to fuzzies. A fuzzy is a young mouse with its first fur; a peach fuzzy sits between |

Past the hatchling stage, size prey by proportion: **a prey item slightly thicker than the snake at its thickest point**. A 70 g adult male may settle on a hopper (a young, weaned mouse) or a couple of fuzzies. A large adult female works up to considerably bigger prey.

### How often

| SNAKE                    | FEED EVERY                                                                               |
|--------------------------|------------------------------------------------------------------------------------------|
| Hatchlings and juveniles | <b>5 to 7 days</b> , or simply weekly                                                    |
| Adult males              | About <b>every two weeks</b> , the default for any adult. Metabolism slows with maturity |
| Mature females           | <b>10 to 14 days</b> ; some large females 14 to 21                                       |

#### THE MOST COMMON FEEDING MISTAKE WITH THIS SPECIES

Keeping an adult on a hatchling's schedule. An adult fed weekly gets fat, and obesity is a real contributor to shortened lifespans in hognoses. Match the frequency and the portion to the adult's activity and size, not to the schedule that suited it as a fast-growing juvenile. Diets high in protein may also predispose a reptile to uric acid building up in the body, which can cause gout, a painful disease of the joints and organs.

### Keep a feeding record

Write down every meal: the date, the prey, and whether it was taken. Weigh the snake once a week on the same scale. A hognose that is getting sick often shows it on the scale before it shows it anywhere else, and a feeding record is what turns "it seems off" into something a vet can use. The Owner Log on page 40 is set up for exactly this.

## Frozen prey: thawing, storage & where to feed

The part most keepers get sloppy on, and it matters more than the schedule.

### Thawing

- **In the refrigerator** until fully thawed, the standard method. Or seal it in a bag and submerge it in cool to lukewarm water, changing the water as needed, until there is no frozen core left.
- **Never on a counter** at room temperature for hours, and **never in a microwave**. A microwave heats unevenly, leaving a frozen center beside hot pockets that can burn the snake's mouth or make the prey burst.
- **Offer it at room temperature or a little warmer**, never cold from the refrigerator and never hot. Warm it gently, for example by running warm water over the outside of the sealed bag, and check often by touch.
- Offer it from **feeding tongs**, never your fingers. Most bites in this species are feeding mistakes, page 16.

### Storage

- Food held at a steady 0°F (−18°C) stays safe indefinitely, but quality declines the longer it sits. Keep prey sealed and airtight, and do not let stock sit for a year or more.
- **Thaw once**. A thawed rodent is never refrozen. If the snake will not eat it, it is a loss, not leftovers.

#### FROZEN RODENTS CARRY *SALMONELLA* TOO

Several *Salmonella* outbreaks in people have been tied to frozen feeder rodents. Wash your hands with soap and water right after handling them or anything where they were stored or thawed. Keep them away from where human food is stored, made or eaten, use containers and utensils that never touch human food, never thaw prey in the kitchen sink you use for dishes, and never use a kitchen microwave or kitchen utensils on it.

### Where to feed: a separate container

Hognoses are enthusiastic, somewhat messy eaters and are prone to swallowing substrate along with their food. **Feed in a separate container**, away from the loose substrate this species needs. It is the single most effective way to prevent impaction, and it is worth doing as standard practice rather than only after a problem.

| STEP                | HOW                                                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The container       | A plain lidded plastic tub with air holes, lined with paper towel. Plain, lined, boring, easy to clean: the one place the setup should look nothing like the display enclosure     |
| Moving the snake in | Lift it out of the substrate with a snake hook, never a bare hand that smells of mouse. Most bites happen with a hungry snake, or while it is being moved into a feeding container |
| Offering the prey   | From tongs, never fingers                                                                                                                                                          |
| Moving it back      | Once it has swallowed, move it back gently and briefly, by hook or by letting it crawl, with no handling session. Then leave it alone for 48 hours                                 |

If the snake refuses, throw the prey out rather than saving it, since thawed prey is never refrozen, and work through the refusal checklist that follows.

## The refusing hognose

This species has a reputation for turning down food. A refusal is usually a puzzle, not an emergency.

A hatchling that turns down a meal is not automatically in trouble, and neither is an adult that skips a few. Work through the causes in order: first the enclosure, then the season and the snake's own cycle, then the prey itself. A refusal becomes a vet question when it comes with weight loss or any other sign of illness. A hatchling or juvenile has far less in reserve than an adult, so weigh a young snake every week through any refusal and call the vet sooner.

### Causes and what to do

| CAUSE                  | HOW YOU CAN TELL                                                     | DO                                                 |
|------------------------|----------------------------------------------------------------------|----------------------------------------------------|
| Enclosure too cool     | Read the thermometers; check the probe                               | Fix the gradient on page 7 before anything else    |
| New arrival            | Home for less than two weeks                                         | Leave it alone to settle, then try again           |
| A shed coming          | Colors fade, eyes cloudy blue, hides more, more defensive            | Offer again after the shed. Page 29                |
| Breeding season        | An adult male ready to breed may refuse                              | Not fixed by offering more food. Weigh it and wait |
| Seasonal slowdown      | Late fall or winter; prefers the cool end, may stay buried for weeks | Normal while its weight holds. Page 21             |
| After a regurgitation  | A meal brought back up in the last two weeks                         | No food for two weeks, then a small meal. Page 27  |
| Picky about plain mice | A healthy, settled hatchling ignores a plain pinky                   | Scent the prey, below                              |

### Scenting

One practical threshold: **start scenting if a snake refuses its first three meal offerings**. Scent the thawed prey with one of these:

- **Fish:** sardine, tuna or salmon.
- **Chicken bone broth.**
- **Frog, toad, lizard or gecko scent**, or commercial scent strips, the packaged option. The amphibian scents work because they match what the species evolved eating.

Change one thing at a time and keep the feeding record, so you know what worked.

#### WHEN A REFUSAL BECOMES A VET VISIT

**Weight down more than 10%** from the snake's starting weight, or visibly thinner along the spine and tail base. **Wheezing, open-mouth breathing**, or bubbles or discharge at the nose or mouth. **Excess saliva, tiny red spots on the gums**, or cheesy yellow material in the mouth. **Lethargy** well beyond the season's usual quiet. Still refusing by spring, or refusing out of season while losing weight. Any of these ends the wait.

#### REFUSED OR REGURGITATED?

A refused meal is a snake that never struck. A regurgitated meal was eaten and came back up. The first usually needs patience. The second means something in the setup, the handling or the snake's health is wrong, and it has its own plan.

## Handling & hygiene

Short, calm, supported sessions, never near a meal, and soap and water every time.

Give a new hognose **about two weeks to settle** before the first session. After that:

1. **Signal that this is not food.** Lift the snake out of the substrate with a small tool such as a snake hook, then move it onto your hands.
2. **Come in from the side.** A hand or tool dropping from above looks like a predator and sets off the hood and hiss.
3. **Support as much of the body as you can.** Let it rest along your palms and forearm. Never grab the tail and never restrain the head.
4. **Let it move.** Keep your hands under it as it crawls, moving one hand ahead of the other.

### How long and how often

| RULE                | DETAIL                                                                                  |
|---------------------|-----------------------------------------------------------------------------------------|
| First sessions      | No more than 5 minutes, and end on a calm snake, not mid-display                        |
| Building up         | To 10 minutes, then to as much as half an hour. Never past an hour                      |
| How often           | Once or twice a week is plenty, and never more than once a day                          |
| Skip a session when | It has eaten in the last 48 hours; it is in a shed (eyes cloudy or blue); it is mid-dig |

### *Salmonella* and the household

Every reptile can carry *Salmonella*, bacteria that live in the gut of a perfectly healthy snake with no sign of illness and leave in its droppings. A single negative test is not a clean bill of health, so treat every reptile as a carrier. The risk is not the bite. It is hands and surfaces to mouth: the snake, its water, its substrate and its gear all carry it.

- **Wash your hands with soap and running water** right after touching the snake, the enclosure, the water or anything that has touched them.
- **Keep the snake and its gear out of the kitchen**, and never clean the enclosure, dish or tools in a kitchen sink, a bathroom sink or a bathtub people use. A dedicated tub or bin solves this.
- **Don't kiss the snake** or hold it close to your face.
- **Children under 5 should not handle reptiles or touch their enclosures at all.** Adults over 65, pregnant people and anyone with a weakened immune system are also at higher risk of serious illness.

#### IF SOMEONE IN THE HOUSE GETS SICK

Diarrhea, fever and stomach cramps are the ordinary symptoms, which is why the link to a pet gets missed. Mention the snake to the doctor even if nobody remembers touching it, especially for anyone in a higher-risk group.

## Hooding, hissing & playing dead

One of the most dramatic defensive acts of any pet snake, and almost all of it is bluff.

When a hognose feels threatened it puts on a show, usually in this order. It inflates its body with deep breaths and **flattens its neck into a cobra-like hood**, the source of its "puff adder" nickname. It **hisses loudly**. It throws **closed-mouth mock strikes** that never make contact. If none of that works, some hognoses go further and **play dead**: they roll onto the back, writhe, open the mouth and let the tongue hang out, and sometimes release musk, a foul-smelling fluid, or feces. Flip one back over mid-performance and it will often roll itself belly-up again, as if it has not noticed. Playing dead is called death feigning, or thanatosis.

### WHAT IT MEANS

A hognose putting on the full display is doing everything it can to avoid biting you. Real aggression in this species is close to unheard of. The act is hardwired, not illness and not a personality, and it is not a sign the snake is about to bite. Bites come from a different place entirely: food.

### What the research adds

A field study of 27 wild western hognoses, published in the journal *Ethology* by Durso and Mullin, simulated predator attacks and timed the response. Larger snakes played dead for longer than smaller ones, and that link was stronger in females than in males. Snakes found at higher temperatures started playing dead sooner. For a keeper, that means a big female may hold the act a long time, and that is still normal.

### What to do when it happens

- **Stay calm and keep your hands steady.** Panicking and dropping the snake is the classic first-timer mistake. Calm, steady handling works better.
- **If it rolls over, stop.** Put it down somewhere safe, step back, and let it right itself on its own. Do not keep flipping it.
- **Approach from the side next time.** A hand or tool dropping from above looks like a predator and sets off the act. Page 14.

### Normal, or worth a look

| NORMAL                                                                            | WORTH A LOOK                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hooding and hissing when startled or disturbed, then settling                     | Hooding and hissing at every routine interaction, for weeks. A snake doing this constantly is reporting something about its enclosure: more cover and deeper substrate at both ends usually reduce it. |
| Rolling over and lying limp, then righting itself once left alone in a quiet room | Limp or unresponsive with no trigger, or still limp long after it has been left alone at normal temperature. That is not the act: call a vet the same day                                              |
| More defensive in the days before a shed                                          | A snake that stays defensive and stops eating outside a shed. Check the setup first                                                                                                                    |

## The venom question & bites

Mildly venomous, not dangerous, and worth respecting at feeding time.

The western hognose is **rear-fanged and mildly venomous**. It has enlarged, ungrooved fangs toward the back of the upper jaw, connected to a gland in front of them called the Duvernoy's gland, which produces a weak venom evolved to help it subdue toads. Bites are uncommon and their effects stay local, though a snake that holds on can cause real swelling and bruising. A 2026 study calls it the most commonly kept venomous pet snake in the world.

### What the research says

| STUDY                                                                                                       | WHAT IT FOUND                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Weinstein and Keyler, 2009</b> , a case report in the journal <i>Toxicon</i>                             | A keeper was bitten while offering a young mouse to a captive snake, which held on to the inside of the elbow for several minutes. The result was marked local swelling, bruising, swollen lymph nodes (the small glands that fight infection), skin signs of a mild cellulitis (an inflamed skin infection) and blisters. There were no effects anywhere else in the body, and recovery was complete after about five months. The authors concluded that hognoses may uncommonly produce medically significant bites but should not be considered dangerous, and that captive snakes should be handled carefully, particularly when offered food      |
| <b>Damm and colleagues, 2026</b> , a survey of keepers in <i>Frontiers in Amphibian and Reptile Science</i> | 63 bite reports, 34 of them with symptoms. 24 of the 63 happened during feeding, and most involved a hungry snake or one being moved into a feeding container. Quick strike-and-release bites caused no symptoms 37% of the time (14 of 38), and the rest cleared within 2 days. Bites where the snake held on or chewed produced symptoms 80% of the time, lasting on average 3 days for a hold and 6 for a chew. Local swelling was by far the most common symptom. 7 of the 63 sought medical help. The authors judged the species relatively harmless, unable to cause permanent damage or death, and in need of no legal regulation for its venom |

### How bites happen, and how to avoid them

Real bites are rare and are almost always a **feeding response**: the snake mistakes a hand for food. So: **wash prey scent off your hands** before handling; **lift the snake out with a snake hook** rather than a bare hand, especially on feeding day; offer food from tongs; and **leave it alone for 48 hours after a meal**. The survey's authors add that proper equipment, leather gloves or a snake hook, and getting the snake off quickly would prevent virtually all envenoming, the term for venom actually entering the body.

#### IF A BITE HAPPENS AND THE SNAKE HANGS ON

Get it off quickly, because the longer a hognose holds and chews, the more symptoms follow. But do not yank it off, which can injure both of you. **Pour cool water over its face** and it will usually let go on its own. Then show a doctor any bite that breaks the skin: tell them it was a rear-fanged hognose. Rare allergic reactions are possible, as with any bite, so get emergency care for any reaction beyond the bite site itself.

None of this puts the western hognose in the category state venomous-reptile laws are written for: several of those laws exempt hognoses by name. The one place the venom does decide the answer is New York City. Pages 22 to 24.

## Enrichment & common mistakes

Researchers let western hognoses choose. They chose enrichment, and chose it more strongly as time went on.

Nagabaskaran, Skinner and Miller ran a preference test, published in the journal *Animals* in 2022. Each enclosure was divided in two. The enriched half had burrowing substrate, objects to interact with, and a large water dish. The standard half had paper towel and a small dish. **Both halves had a shelter, and there was a heat gradient**, so the test compared enrichment on top of adequate care, not enrichment against neglect. Over six days of observation:

- The snakes **preferred the enriched side**, and the preference **grew stronger over time**, which rules out simple curiosity about something new.
- The preference was **stronger when the enriched side was the cooler one**, which the authors link to this species' burrowing.
- Boldness made no difference. A shy hognose is not opting out of enrichment.

### Priority order

1. Substrate deep enough to burrow into completely
2. That deep substrate placed on the cool side
3. A water dish large enough to soak in
4. Secure cover at both ends of the gradient
5. Scent novelty and varied feeding presentation
6. Rearranging the furnishings, and new objects, last

### Common mistakes

| MISTAKE                                                        | WHY IT MATTERS                                                                                                               |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Misting, or a damp enclosure                                   | The root of respiratory infection and scale rot. Page 9                                                                      |
| Paper towel as the permanent floor                             | It is the exact condition the study's standard side represented. Paper towel is right for quarantine, page 18, and not after |
| Deep substrate on the warm end                                 | It pits two preferences against each other                                                                                   |
| Feeding on the substrate                                       | Impaction. Page 12                                                                                                           |
| An adult on a hatchling's schedule                             | Obesity and a shorter life. Page 11                                                                                          |
| A heat source with no thermostat                               | Burns. Page 8                                                                                                                |
| Reading the hood as personality                                | It is a threat response. Page 15                                                                                             |
| Handling a snake that is mid-dig, or within 48 hours of a meal | It interrupts the behavior the enclosure exists for, and the second invites a feeding bite                                   |
| Collecting a wild hognose                                      | Illegal in many states, and an eastern or southern hognose is often protected. Page 22                                       |

## Choosing a hognose, quarantine & the first vet visit

Captive-bred, eating frozen-thawed mice, and kept apart for months. In that order.

Buy **captive-bred**. Compared with wild-caught animals, captive-bred reptiles are less likely to bite, more likely to accept the food offered, and less likely to carry parasites. A breeder who hatched the snake can tell you when it hatched, what it eats, and when it last ate and shed. A shop or breeder is worth buying from only if its enclosures are clean and large enough. A rescue snake often arrives with no reliable history, and an expo puts animals from many sellers in one room for a day. Neither is a reason to refuse the animal. Both are reasons to run the full quarantine below.

### Look before you pay

| CHECK                                  | HEALTHY                                  | WALK AWAY OR ASK MORE                                                           |
|----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------|
| Eyes                                   | Clear and alert                          | Swollen, sunken, crusted, or cloudy outside a shed                              |
| Nose, breathing                        | Dry nostrils, quiet breathing            | Bubbles, mucus, wheezing, open-mouth breathing                                  |
| Vent (the rear opening under the tail) | Clean                                    | Stained, pasted with feces, or swollen                                          |
| Body                                   | Rounded, no ridge of spine showing       | Visible bones, or well behind its age in size                                   |
| Skin and shed                          | Intact, no old shed left on              | Wounds, or rings of old shed on the tail tip                                    |
| Mites                                  | None visible                             | Tiny black, brown, red or orange specks, especially around the eyes             |
| Behavior                               | Tongue flicking, aware, good muscle tone | Limp or unresponsive. A hood and a hiss at a stranger's hand is normal, page 15 |

**Ask the seller:** the hatch date; whether it was captive-bred; what it eats, when it last ate, and whether it takes plain or scented frozen-thawed mice; whether it has been treated for anything, mites or a respiratory infection above all; and for a receipt naming *Heterodon nasicus*. A good breeder answers without looking anything up.

### Quarantine: 3 to 6 months

Quarantine means keeping a new animal completely apart from any other reptile. A new reptile needs **3 to 6 months** of it. If this is your only reptile, the separation rules below have nothing to protect, but the first vet visit and its fecal test still apply.

| RULE              | HOW                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------|
| A separate room   | A second enclosure on the same shelf shares the same air. A different room is the real separation                            |
| Paper towel floor | It shows mites, stool and regurgitated food at a glance. Just a hide and a water dish, things you can disinfect or throw out |
| Separate tools    | Tongs, dish, scoop, even the paper towel roll                                                                                |
| New animal last   | Care for established animals first, every time. Wash your hands, or change gloves, between                                   |
| Restart the clock | If anything flags, a positive fecal test, a regurgitation, unexplained weight loss, start the window again                   |

#### FIND THE VET BEFORE THE SNAKE COMES HOME

Not every practice treats reptiles. The Association of Reptile and Amphibian Veterinarians runs a find-a-vet directory. Ask a clinic how many reptiles it sees, whether it treats snakes routinely, whether it runs fecal tests (stool checks for parasites) in house, and who covers emergencies after hours. Book the first exam within 72 hours of bringing the snake home, and no later than two weeks, with a fresh stool sample in a sealed bag. That visit records a baseline weight and the snake healthy.

## Telling the sex, weight & body condition

Sex is a vet's job. Weight and body condition are yours, every week.

### Telling male from female

Snakes show no outside sign of sex that an owner can trust. Two hands-on methods work, and both are procedures with real technique behind them:

| METHOD            | HOW IT WORKS                                                                                                                                                                                                                                               | WHO SHOULD DO IT                                                                     |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Probing</b>    | A lubricated, blunt metal probe slides into the vent and back along the tail. In a male it goes deeper, into the pocket holding a hemipenis (one of a male snake's paired reproductive organs), about 6 to 14 scales; in a female it stops at about 3 to 6 | A reptile vet or experienced breeder only. The wrong size or angle injures the snake |
| <b>Popping</b>    | Gentle pressure at the base of a hatchling's tail turns the hemipenes outward in a male                                                                                                                                                                    | A reptile vet or experienced breeder only. Too much force injures the tail           |
| <b>Ultrasound</b> | Imaging that shows the internal organs. Reliable in snakes and needs no handling of the vent                                                                                                                                                               | A reptile vet                                                                        |

Tail shape is a weak guess on one animal with nothing to compare it against. In an adult hognose, size is a strong hint, because females run so much larger, but it is a hint, not confirmation.

### Weigh it every week

Weight, not length, is the number that tells you how a snake is doing. Weigh it once a week on a kitchen scale in grams: set a lidded tub on the scale, zero it, then add the snake. One reading tells you little. Three or four in a row, climbing, holding or falling, tell you a great deal, and it is the direction of that line that should drive a feeding change or a vet call. Use page 5 for rough adult weights, and your own snake's record for everything else.

#### THE WEIGHT LINE TO WATCH

A drop of more than 10% from the snake's normal weight, or a steady fall across several weeks, is a vet visit, whatever the season. A sudden drop is the first thing to show the vet.

### Body condition: the cross-section test

Support the snake and look at it end-on, picturing a slice straight across its body.

| SHAPE            | WHAT YOU SEE AND FEEL                                                                                       | DO                                                                                                                   |
|------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Right</b>     | A rounded loaf, full and even, no sharp angles. Muscle over bone along the back                             | Keep going                                                                                                           |
| <b>Too thin</b>  | Triangular, sloping away from a ridge down the middle of the back where the spine sits close to the surface | Over several checks, and with falling weight, a vet visit: thinness is as often parasites or illness as underfeeding |
| <b>Too heavy</b> | Round in a different way: skin folds along the sides, or a soft, doughy feel over the ribs                  | Feed smaller or less often, page 11. Do not stop feeding outright                                                    |

The 9-point body condition scale used for dogs and cats does not apply to a snake's body. The cross-section check is the one that fits, and it is the one a reptile vet uses.

## Females, eggs & eggs that get stuck

For anyone keeping a female that has been with a male, or bought from a breeder who paired her.

This page is not a breeding guide. It is what a keeper of a female needs to recognize, so that a normal clutch is handled calmly and a stuck one gets to a vet in time. A female is **gravid** when she is carrying eggs.

### What a normal clutch looks like

| STAGE            | WHAT HAPPENS                                                                                                       |
|------------------|--------------------------------------------------------------------------------------------------------------------|
| Mating           | In the wild, June to August. Wait until a female is at least 2 years old, preferably closer to 3, and 150 to 200 g |
| The pre-lay shed | She sheds, then lays on average <b>7 to 11 days later</b> . That shed is your signal to put the laying box in      |
| The clutch       | 4 to 23 eggs in a wild female, around 13 on average. Captive clutches have topped 30                               |
| Hatching         | 52 to 64 days in the wild. In an incubator at 78 to 82°F (25.6 to 27.8°C), 51 to 60 days                           |

### The laying box

A plastic food container with a hole cut in the lid, filled about halfway with loamy soil. Put it in after the pre-lay shed and leave her alone. Solitude and a suitable place to lay are what she needs, and an inadequate nest site is one of the causes of eggs getting stuck.

### Egg binding

Egg binding, also called dystocia, is when a female cannot pass her eggs. It is life-threatening.

| EGG BINDING      |                                                                                                                                                                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>     | Poor husbandry, an inadequate nest site, an inappropriate diet, and dehydration                                                                                                                                                               |
| <b>Signs</b>     | A female past her expected laying window who becomes lethargic, weak or unresponsive. It is not usually sudden: reptiles can hold eggs for weeks or even months, so a female that has not laid when expected needs watching closely           |
| <b>Diagnosis</b> | The vet examines her and uses X-ray images and ultrasound to find the eggs                                                                                                                                                                    |
| <b>Response</b>  | Immediate vet evaluation. The vet may try medical treatment and better conditions, but medical management often fails and surgery is needed in most cases. Never give any medication at home: the vet chooses the treatment and sets any dose |
| <b>Recovery</b>  | Depends on how long the eggs were held and what the surgery involved. She rests, and the vet decides when feeding restarts                                                                                                                    |

#### A FEMALE KEPT ALONE

Whether a female hognose kept on her own will produce eggs is not well documented. If your adult female swells, sheds and then turns restless or stops eating with no male in the picture, describe it to your vet rather than guessing.

## The winter slowdown

A hognose may go off food and stay buried for weeks in winter. The question is never the appetite alone.

In the wild a western hognose spends roughly September to March below ground. Brumation is the reptile version of that seasonal slowdown, tied to temperature and daylight: less activity, less appetite, a slower metabolism. It is not the same as a mammal's hibernation, and a brumating snake can still wake, move and drink.

**A pet hognose kept at steady temperatures needs no deliberate brumation.** That is a breeder's tool for bringing on reproduction. But a pet may slow down on its own in late fall even with nothing changed: less active, buried for weeks, off food, and preferring the cool end. That is fine while its weight holds. The trouble is that illness also starts with a refused meal, so check the rest of the snake.

| CHECK                      | NORMAL SEASONAL SLOWDOWN                                             | ILLNESS: CALL A REPTILE VET                                               |
|----------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------|
| Temperatures               | Unchanged, verified at both ends                                     | A failed heater or thermostat explains the fast. Fix it first, then watch |
| Weight, every 1 to 2 weeks | Stable, or a slow drift well inside 10% of the starting weight       | Down more than 10%, or visibly thinner along the spine and tail base      |
| Breathing                  | Quiet, mouth closed                                                  | Open-mouth breathing, bubbles or discharge at the nose or mouth, wheezing |
| Mouth                      | Pink, clean gums                                                     | Excess saliva, tiny red spots on the gums, cheesy yellow material         |
| Behavior                   | Hides more and prefers the cool end, but moves normally when handled | Lethargy well beyond the usual slowdown                                   |
| Timing                     | Starts in fall, ends by spring                                       | Still refusing food by spring, or refusing out of season                  |

A snake that passes every line can be left alone with fresh water available and weighed every one to two weeks. To weigh a snake that is buried, lift it gently and weigh it in a lidded tub.

### Deliberate brumation is an advanced technique

Some keepers cool a western hognose on purpose: stopping food at the beginning of November so the gut empties, then about 3 months of cooling from December to the end of February. It is for an experienced keeper, usually breeding, and only for a healthy snake at a good weight with a clean bill of health from a reptile vet, since cooling an underweight or parasite-carrying snake adds real risk. Lower and raise temperatures gradually over a week or two, keep water available throughout, and offer the first meal only once the snake has been back at normal temperatures for several days and is active.

#### STOP AND WARM IT BACK UP IF

It loses more than 10% of its starting weight, shows any sign of respiratory illness (wheezing, mucus, open-mouth breathing), is lethargic well beyond the expected slowdown, or does not wake as expected. Then call the vet.

## Is a hognose legal where you live?

Broadly legal. The exceptions turn on native-species law far more often than on venom, and on which hognose you have.

This section covers all 50 states, Washington, D.C. (the District of Columbia) and New York City, which has its own rules. Of those 52, a pet hognose is **legal in 27, conditional in 15, permit-only in 4, and banned in 6**. The note beside each says which hognose the rule reaches, because in several states that differs between the western and the eastern. **Three of the six bans, Georgia, New York State and West Virginia, reach only their native hognoses, not the western.**

|                    |                                                                                          |               |                                                     |
|--------------------|------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------|
| <b>Legal</b>       | The state's own rules allow a pet hognose with no permit                                 | <b>Permit</b> | Legal, but a permit or license comes first          |
| <b>Conditional</b> | Legal without a permit only if a stated condition is met, very often which species it is | <b>Banned</b> | Cannot be kept as a pet, and no permit route exists |

### THE VENOM QUESTION, LEGALLY

The mild venom almost never decides the answer. Several state venomous-reptile laws name hognoses as an exception: Kentucky and Alabama write "except for hognose snakes (genus *Heterodon*)" into the rule itself. What catches hognoses is native-wildlife law. The eastern and southern hognoses are native and often protected, and a few states protect the western as their own native too. The one place the venom itself decides it is New York City.

## Banned

| WHERE                 | WHY                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Georgia</b>        | Both Georgia hognoses, the eastern and the southern, are on the Department of Natural Resources list of native species that "may not be held as a pet regardless of its origin or morphology," so a captive-bred animal or a color form is caught too. The western is not a Georgia species and is not on that list, so which hognose you have decides it. Check with the department before keeping any hognose |
| <b>Hawaii</b>         | Every snake is banned, with no exception for harmless species                                                                                                                                                                                                                                                                                                                                                   |
| <b>Kansas</b>         | Both Kansas hognoses, the western (named in the rule as the plains hog-nosed snake) and the eastern, are listed as nongame species in need of conservation, and no rule or permit allows keeping one. An animal held before the rule may be kept only if an affidavit was filed and approved before 1 January 1990                                                                                              |
| <b>New York State</b> | The eastern hognose is a protected native snake with no open season and a species of special concern, and a captive-bred animal from another state is treated the same. The only possession license is not issued for pets. The western is not a New York species and the state rule reaches only the eastern, so which hognose decides it. Confirm with the Department of Environmental Conservation           |
| <b>New York City</b>  | The city health code bans every venomous, rear-fanged colubrid (a member of the snake family Colubridae) with a Duvernoy's gland, "even if de venomized." That reaches the western hognose. A prohibited animal can be seized                                                                                                                                                                                   |
| <b>West Virginia</b>  | The state bars keeping any native reptile not on a short permitted list, which names no snake. The eastern hog-nosed snake is on the state's checklist of native snakes                                                                                                                                                                                                                                         |

## Permit

| WHERE             | WHY                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Illinois</b>   | The western hognose is listed as a threatened species in Illinois, so keeping one needs a Department of Natural Resources threatened species possession permit, and breeding or selling one needs a separate propagation permit at \$150 a year. The eastern is not listed and falls under the ordinary cap of 4 per species and 8 in total. Cities can be stricter |
| <b>Missouri</b>   | Both the eastern and the western are Missouri natives and are not treated as venomous, which puts them in Class I of the state Wildlife Code. Confirm what that requires with the Department of Conservation                                                                                                                                                        |
| <b>New Jersey</b> | Both need a permit: the eastern as a native nongame species, the western as an exotic reptile. Taking an eastern from the wild is separately illegal, so a permitted one has to be captive-bred                                                                                                                                                                     |
| <b>New Mexico</b> | All three hognoses are on the importation list in Group III, a permit with specific conditions attached. The rule covers bringing an animal into the state rather than keeping one already there                                                                                                                                                                    |

## States with conditions

Fifteen states where a hognose is legal only on a stated condition. Read the condition, then check which species you have.

| WHERE                | THE CONDITION                                                                                                                                                                                                                                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Alabama</b>       | Only the southern hognose is protected, and keeping one needs a scientific collection permit or a written permit from the Commissioner. The eastern and the western are unrestricted, and the state's venomous-reptile ban names hognoses as an exception                                                                         |
| <b>Colorado</b>      | A captive-bred western hognose is allowed with documentation and no license, under the rule in effect from 1 May 2026. The eastern is excluded by name. A hognose taken from the wild in Colorado is lawful only if held on or before 1 May 2026 and reported by 30 July 2026; no new wild capture. Check your city or county too |
| <b>Connecticut</b>   | The eastern is a State Species of Special Concern and needs a permit to import or possess. The western is not a Connecticut species and is unrestricted                                                                                                                                                                           |
| <b>Delaware</b>      | The eastern is native and not available for private ownership, except a captive-bred animal in an unusual color form held under permit. The western, not a Delaware species, needs an Individual permit from the Department of Agriculture                                                                                        |
| <b>Indiana</b>       | The eastern is an Indiana native: selling, offering or transporting one for sale is barred whatever its origin, and keeping is capped at 4 of a native species. Captive-bred albino, leucistic (pale) or xanthic (yellow) color forms are exempt, and a captive breeding license allows sales. The western is not on the list     |
| <b>Maryland</b>      | The eastern: one per person without a permit. The western is outside the rule and unrestricted. Counties can be stricter, so check the local rule first                                                                                                                                                                           |
| <b>Massachusetts</b> | The western is exempt by name. The eastern is a state species of special concern, which puts it in a permit tier the state does not issue for pets, captive-bred or not                                                                                                                                                           |
| <b>Michigan</b>      | The eastern is native and may be taken only with a fishing license, at limits of 3 a day and 6 in total across snakes, lizards and turtles, and reptiles may not be bought or sold. The western is not a Michigan species and is legal. Local rules can be stricter                                                               |
| <b>Minnesota</b>     | A snake from a permitted breeder, or one held before 1 August 2021, may be kept as a pet. Both hognoses are Minnesota natives. The breeder permit system is not visibly in place yet, so ask the Department of Natural Resources how a buyer meets the condition                                                                  |
| <b>Nebraska</b>      | Both hognoses are capped at 3 per species for collected animals. One legally bought from outside Nebraska may be kept with no number limit, if you can show proof of legal purchase such as a bill of sale. Keep the receipt                                                                                                      |
| <b>Oklahoma</b>      | Both hognoses reach Oklahoma. A hunting license is required to take or possess a land reptile, with a limit of 6 per species, and state law exempts a legally obtained pet from a source other than the wild from the possession ban. Ask the Department of Wildlife Conservation which applies to your snake                     |
| <b>Pennsylvania</b>  | The eastern is native, with no open season and a limit of zero, and cannot be imported. The western is outside the rule, so a captive-bred western from a shop is lawful                                                                                                                                                          |
| <b>Texas</b>         | Both hognoses are on the state nongame list: up to 25 per species without a permit, and no selling, trading or bartering any of them. This applies to captive-bred animals too                                                                                                                                                    |
| <b>Virginia</b>      | The eastern is a Species of Greatest Conservation Need and may not be possessed for private use in any number. The western is a nonnative animal and needs no permit                                                                                                                                                              |
| <b>Wisconsin</b>     | The eastern is native, capped at 5 until the end of 2026. From 1 January 2027 only captive-bred color variants from a legal source may be kept, and a wild-type eastern already held must be registered with the Department of Natural Resources by 30 June 2027. The western is not a Wisconsin species                          |

## States where it is legal

Twenty-seven jurisdictions with no state permit for a pet hognose. A few still have a catch.

| WHERE                                                                                                                | NOTE                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Maine, New Hampshire, Oregon, Rhode Island</b>                                                                    | Each names the <b>western</b> hognose specifically. The eastern is a different answer: in Maine it is a native left off the list; in New Hampshire it is controlled separately; in Oregon it is on no list and may not be possessed; in Rhode Island keeping one as a pet is prohibited. Oregon cities and counties can be stricter. Maine's list was last verified in August 2026 |
| <b>Kentucky</b>                                                                                                      | The venomous-snake ban lists the colubrid family and then carves hognoses straight back out by name. Local rules can be stricter                                                                                                                                                                                                                                                   |
| <b>Florida</b>                                                                                                       | Cleared as a nonvenomous, unprotected reptile. Neither Florida hognose is on the state's imperiled species list                                                                                                                                                                                                                                                                    |
| <b>California, Montana, North Carolina, Ohio, Washington</b>                                                         | Legal at state level, but cities and counties may prohibit what the state allows. Check your local ordinance (a city or county law). In Ohio the eastern is native, and collecting one is a wildlife agency question                                                                                                                                                               |
| <b>Nevada</b>                                                                                                        | Cleared as a nonvenomous reptile not native to Nevada. Clark County, Las Vegas and Henderson are stricter than the state: check the city first                                                                                                                                                                                                                                     |
| <b>Arizona, Wyoming</b>                                                                                              | Legal to keep a captive-bred animal. Hognoses live wild in both states, and taking one from the wild is a separate question with its own license and limits. The Mexican hognose reaches southeastern Arizona                                                                                                                                                                      |
| <b>South Dakota</b>                                                                                                  | No possession permit, but a free entry permit applies to bringing any nondomestic animal into the state                                                                                                                                                                                                                                                                            |
| <b>District of Columbia</b>                                                                                          | Its short list of allowed pets includes nonvenomous snakes, and the hognose reads as permitted under it                                                                                                                                                                                                                                                                            |
| <b>Alaska, Arkansas, Idaho, Iowa, Louisiana, Mississippi, North Dakota, South Carolina, Tennessee, Utah, Vermont</b> | No state rule restricts a captive-bred pet hognose                                                                                                                                                                                                                                                                                                                                 |

### Before you buy, wherever you live

- **Know the species.** Get *Heterodon nasicus* and "captive-bred" on the receipt. In most of the conditional states, that line is the condition.
- **Never collect a wild hognose.** Wild collection is restricted or banned in many states that allow captive-bred pets, and the eastern and southern hognoses are protected in several.
- **Check local rules.** A city or county ordinance can be stricter than the state, and no state list covers it.
- **Moving?** Check the new state before you go. A legal pet in one state can be capped, permit-only or banned in the next.
- **Keep the paperwork:** the receipt, any permit, and the breeder's details. Nebraska and Colorado both turn on proof of legal origin.

#### NOT LEGAL ADVICE

This section is a summary of state rules as of October 2026, drawn from each state's regulations and agency pages, last checked on 1 October 2026 for every jurisdiction except Maine. It is not legal advice. Rules change, sometimes on a fixed date, as Wisconsin's will on 1 January 2027. Before you buy, check with your state wildlife agency, and with your city or county.

# Health red flags & finding a vet

Western hognoses are hardy. Most of what goes wrong traces back to humidity, substrate or feeding, all within your control.

## CONTACT A REPTILE VET PROMPTLY IF YOU SEE

Open-mouth breathing, wheezing, drooling or mucus · no droppings with bloating and lethargy · belly scales discolored or blistered and getting worse · red spots on the gums, excess saliva or cheesy material in the mouth · mites · a burn · anything protruding from the vent · a meal brought back up twice · weight down more than 10% · limp or unresponsive at normal temperature · an eye cap still on after a shed · a female past her laying window who is weak or listless

## What to tell the vet

- ☐ Basking, cool-side and night temperatures, and what controls the heat
- ☐ Humidity, and where you measured it
- ☐ The last meal: date, prey, and whether it stayed down
- ☐ When the signs started and whether they are getting worse
- ☐ The last normal dropping, and what it looked like
- ☐ The weight record from page 40, and a fresh stool sample in a sealed bag

## Why the setup comes up first

Two of the most common problems in this species, respiratory infection and scale rot, share one cause: humidity or dampness running too high. Feeding on loose substrate causes impaction. Overfeeding against the snake's activity causes obesity. So a reptile vet will ask about your setup before anything else, and having the numbers ready shortens the visit. Snakes also hide illness well, which is why a weekly weight and a written record of meals, sheds and behavior catch problems before they show.

## What a vet visit costs

|                                                   |                           |
|---------------------------------------------------|---------------------------|
| Routine wellness exam                             | \$50 to \$100             |
| Fecal test (a stool check for parasites)          | \$25 to \$50              |
| Radiograph, if needed                             | \$100 to \$250 per view   |
| Treating a respiratory infection, or an impaction | Can run into the hundreds |

## Normal or concerning

| NORMAL                                                                               | CONCERNING                                                             |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Hooding, hissing or playing dead when disturbed                                      | Limp or unresponsive with no trigger. Page 15                          |
| Refusing a meal or two in good body condition, especially in winter or before a shed | Refusal with weight loss, or with any other sign on this page. Page 13 |
| Cloudy, blue eyes and faded color for a few days before a shed                       | Shed coming off in pieces, or an eye cap left on. Page 29              |
| Buried for days at the cool end, weight steady                                       | Buried and losing weight, or wheezing. Page 21                         |
| A semi-formed dropping with fur, and a white urate                                   | Watery or bloody droppings, or yellow or gritty urates. Page 30        |

## Respiratory & belly-scale infections

Two of the most common problems in this species, and the same root cause: too much moisture.

### Respiratory infection

|                  |                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>     | Humidity running too high, temperatures too cool, or poor ventilation. This species prefers a dry setup, and dampness is more of a problem for it than for many other snakes. Unsanitary conditions, other illness and poor nutrition also contribute                                                                                                                                                    |
| <b>Signs</b>     | Open-mouth breathing, wheezing, drooling or visible mucus, discharge from the nose, difficulty breathing, and lethargy                                                                                                                                                                                                                                                                                   |
| <b>Diagnosis</b> | A reptile vet's examination, together with your temperature and humidity readings                                                                                                                                                                                                                                                                                                                        |
| <b>Response</b>  | Always see a vet: it needs antibiotics and will not resolve on its own. The vet chooses the drug and sets the dose; never give leftover or bought antibiotics. At home, fix the cause at the same time: dry the enclosure, clean it, and hold the warm end correctly, because a sick snake is kept toward the upper end of its normal range. A respiratory infection gets worse in a snake kept too cool |
| <b>Recovery</b>  | Depends on how early it is caught and on fixing the setup. If the humidity stays high, expect it to come back                                                                                                                                                                                                                                                                                            |

#### THE BEST TOOL FOR CATCHING IT EARLY

A digital thermometer and hygrometer, because they let you confirm the enclosure is running dry and warm rather than guessing. Page 9.

### Scale rot (blister disease)

|                  |                                                                                                                                                                                                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>     | Damp, dirty substrate                                                                                                                                                                                        |
| <b>Signs</b>     | Discolored or blistered belly scales. Blister disease is the early stage of scale rot: the skin develops pustules (small pus-filled bumps) or blisters                                                       |
| <b>Diagnosis</b> | A vet's examination of the skin                                                                                                                                                                              |
| <b>Response</b>  | Move the snake to a clean, dry setup and keep the substrate dry and clean, which is what this species needs anyway. See a vet if it moves past mild discoloration; any treatment and dose are the vet's call |
| <b>Recovery</b>  | Caught early, the blisters may heal without progressing to open sores. Left alone, they can                                                                                                                  |

#### ONE CAUSE, TWO DISEASES

If you see either of these, read the hygrometer before anything else. A western hognose kept at 30 to 50% humidity, with no misting and a dry, clean bed, is very unlikely to see either one.

## Blockages, regurgitated meals & obesity

Three feeding problems, each with a feeding fix.

### Impaction

**Cause:** a gut blocked by swallowed substrate. Hognoses are enthusiastic, somewhat messy eaters and are prone to swallowing substrate along with their food. **Signs:** a missed stool past its usual gap after a meal, then bloating, lethargy and refused food. The missed stool usually comes first. **Diagnosis:** a vet's examination, often with a radiograph. **Response:** there is no fixed number of days to wait, since the gap depends on meal size and temperature. A missed stool together with bloating, lethargy, refused food or regurgitation needs a reptile vet, not more waiting, and severe cases can need surgery. Prevention is feeding in a separate container, page 12. **Recovery:** depends on the blockage and how it was cleared; afterward, every meal goes in the feeding container.

### Regurgitation

A hognose that brings up a meal undigested or half digested is telling you something went wrong around feeding. **Cause:** handling within 48 hours of a meal, an enclosure running too cool, stress, a meal that was too big, dehydration, a dirty enclosure, intestinal parasites, contaminated prey, or sharing the enclosure with another snake. **Signs:** the meal found in the enclosure or feeding tub. **Diagnosis:** working through that list, and a vet if it repeats, since repeated regurgitation after meals is also the main sign of *Cryptosporidium* (a stomach parasite, next page).

#### AFTER A REGURGITATION

**Leave the snake completely alone:** no handling and no feeding. Check the temperatures. After **two weeks** of steady improvement, offer a small meal, **a quarter to half the usual size**, then rest it again until the next feeding. A snake that looks worse after regurgitating, or brings up that small meal too, goes to a reptile vet.

**Recovery:** most snakes return to normal meals once the cause is fixed. The repeat is what matters, so log every one.

### Obesity and fatty liver

**Cause:** in captivity a hognose gets less activity than in the wild and is sometimes fed too much or too often for its enclosure. The most common version is an adult kept on a hatchling's schedule. It is a real contributor to shortened lifespans in this species, and the serious end of it is fatty liver disease. **Signs:** a round cross-section with skin folds along the sides or a doughy feel over the ribs (page 19), and a weight that keeps climbing in an adult. **Diagnosis:** body condition and the weight record; a vet confirms and checks the liver. **Response:** match feeding frequency and portion to the adult's activity and size, page 11: smaller meals or longer gaps, not a sudden fast. **Recovery:** slow and steady over months, tracked on the scale.

#### GOUT, THE OTHER COST OF OVERFEEDING

Diets high in protein may predispose a reptile to uric acid building up, which can cause gout. Gout can be very painful, to the point that a reptile refuses to move, eat or drink. Yellow or gritty urates are an early hint. Diagnosis and any treatment are the vet's.

## Mites & internal parasites

Both are treatable, and both are why a new snake is quarantined. The third one here is not.

### Snake mites

|                  |                                                                                                                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>     | Tiny blood-feeding parasites, <i>Ophionyssus natricis</i> , most under 1.5 mm (millimeters) long. They spread by direct contact and shared equipment, almost always arriving on a new animal, used decor or tools                                         |
| <b>Signs</b>     | Tiny black, brown or red specks around the eyes, chin and vent, on the shed skin, or moving on the snake or the glass. Rubbing, lethargy, a lost appetite, coarse-looking skin, bad sheds, and long soaking in the water dish, as if trying to drown them |
| <b>Diagnosis</b> | Rub the snake gently over a sheet of white paper: mites that fall off show up against it. The vet confirms                                                                                                                                                |
| <b>Response</b>  | A reptile vet treats the snake and sets the product, dose and schedule. You treat the enclosure the same week: clean it thoroughly, and throw away the substrate, branches and anything disposable. Use paper towel until the vet says it is clear        |
| <b>Recovery</b>  | Good when the snake and the enclosure are both treated. Treat only one and the mites come back. A heavy infestation causes anemia (blood loss) and can carry infections                                                                                   |

#### NEVER TREAT A HOGNOSE FOR MITES ON YOUR OWN

Permethrin-based products (a common insecticide), pest strips and some over-the-counter mite sprays can be highly toxic to hognoses. Veterinary medicine does use a permethrin licensed for reptiles, which is exactly why the choice and the dose belong to the vet, not to a pet store shelf.

### Internal parasites

**Cause:** worms and single-celled parasites, more common in wild-caught animals. **Signs:** weight loss despite eating, repeated watery droppings, regurgitation, or none at all. **Diagnosis:** a fecal test, which is why every new snake has one during quarantine and why periodic checks are worth it after. **Response:** a vet-prescribed antiparasitic drug at a dose the vet sets, plus a full enclosure clean. **Recovery:** usually good; the vet may retest to confirm.

### *Cryptosporidium*

|                  |                                                                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cause</b>     | <i>Cryptosporidium serpentis</i> , a single-celled parasite of the stomach, and a serious concern in this species                                                                                        |
| <b>Signs</b>     | Regurgitation after meals, marked weight loss, slow decline, and often a firm swelling in the stomach region of the body that a vet can feel                                                             |
| <b>Diagnosis</b> | A special stain of fresh droppings, or of the coating on a regurgitated meal                                                                                                                             |
| <b>Response</b>  | Isolate the snake and its tools, and see a vet. Several treatments have been tried and none has been especially effective                                                                                |
| <b>Recovery</b>  | There is no reliable cure. That is the strongest argument on this page for buying captive-bred and running the full quarantine on page 18. The species found in reptiles is not thought to infect people |

## Shedding, mouth infections, burns & tissue at the vent

One normal process to support, and three problems that each need a vet.

### The shed cycle

Young hognoses shed most often. A shed is coming when the colors fade, the eyes turn cloudy blue, and the snake eats less, hides more and turns more defensive. The eyes stay cloudy for about 4 to 7 days, then clear, and the shed follows about 4 to 7 days after that. Leave it alone for that stretch: no handling and no rearranging. Shedding is called ecdysis.

#### HELPING A DRY-COUNTRY SNAKE SHED

Keep the rest of the enclosure dry and add moisture in one spot: mix water into the substrate in the humid hide to help a shed start or finish. Rough surfaces such as cork bark give it something to rub against. After the shed, find the skin and check that **both eye caps and the tail tip** came away with it.

**Stuck shed**, called dysecdysis, has six common causes: low humidity, skin parasites, nutritional problems, infectious disease, nothing rough to rub against, and low thyroid function. A shed that comes off in pieces now and then is not alarming; a pattern of bad sheds in a correct setup is a vet question. **A retained eye cap** (the clear scale over each eye, also called the spectacle) that extra moisture does not loosen needs a vet, and you never remove it yourself: left on, it can cause permanent eye damage and blindness.

### Mouth rot (infectious stomatitis)

**Cause:** infection and inflammation of the tissue lining the mouth. **Signs:** tiny purplish-red spots in the mouth early, then excess saliva and cheesy yellow material along the teeth, and refused meals. **Diagnosis:** a vet's look inside the mouth. **Response:** a vet removes dead tissue, cleans with an antiseptic, and prescribes antibiotics and supportive care, at doses the vet sets. **Recovery:** best when treated early. A snake that is not eating because of it needs the vet's follow-up plan.

### Burns

**Cause:** a snake resting on a spot that is too hot, or staying on a hot spot too long: an exposed lamp or bulb, a heat rock, an unregulated heat source, or a radiator or baseboard heater after an escape. **Signs:** damaged or discolored skin where the snake lay against the heat. **Diagnosis and response:** a vet, every time, plus the thermostat and screen fixes on page 8. **Recovery:** the vet judges it by how deep the burn goes.

### Prolapse

**Cause:** tissue pushed out through the vent, the single rear opening, usually after straining: from impaction or constipation, dehydration, parasites, egg binding or bladder stones. The cloaca (the chamber just inside the vent), a male's hemipenes or part of the colon can be involved. **Signs:** moist tissue visible outside the vent. **Response:** an emergency. Keep the tissue clean and moist with a damp paper towel or clean cloth, **do not push it back in**, and get to a vet immediately: exposed tissue dries and swells fast. **Diagnosis and recovery:** the vet cleans and replaces the tissue and looks for the cause, because without fixing the cause it tends to come back.

## Reading droppings & hydration

The cheapest daily health check there is. Look before you scoop.

A snake does not pass liquid urine the way a mammal does. Its kidneys pull nitrogen waste out of the blood as uric acid, which leaves as **urates**: a white to cream, chalky or toothpaste-like paste. Because a snake has a single rear opening, the cloaca, the urates come out with the stool in the same pass. A snake generally does not pass a dropping until a meal is mostly digested, so the rhythm follows feeding: weekly for a weekly-fed juvenile, much less often for an adult fed every two weeks. Learn your own snake's normal; a real change from it is the signal.

| IT LOOKS LIKE                                                          | USUALLY MEANS                                                                                   | DO                                                                                               |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| A semi-formed dropping with some fur or bone, and a smooth white urate | Normal for a snake eating whole mice                                                            | Nothing. Note the date                                                                           |
| Urate yellow, orange, or hard and gritty                               | Dehydration at least; if it persists, a urate or kidney problem, and the same chemistry as gout | Check the water dish is full and large enough to soak in, and the humid hide. Vet if it persists |
| Watery or unformed, more than once in a row                            | A classic sign of parasitic disease                                                             | Vet, with a fresh sample in a sealed bag. Page 28                                                |
| Bright red blood                                                       | Fresh bleeding near the vent                                                                    | Vet the same day                                                                                 |
| Dark, tarry                                                            | Digested blood from higher in the gut                                                           | Vet the same day                                                                                 |
| Straining, or visible discomfort passing waste                         | Can point to a blockage or a problem at the cloaca                                              | Vet. Page 29 for prolapse                                                                        |
| An unusually strong or off smell                                       | Worth a vet check alongside any other sign                                                      | Note it on the log                                                                               |
| Nothing at all, with bloating and lethargy                             | Possible impaction                                                                              | Page 27                                                                                          |

### Keeping a dry-country snake hydrated

Dry air does not mean a thirsty snake. A **large water dish**, emptied and refilled with fresh water every day, is the hydration plan, and the humid hide is the one damp spot in the enclosure. The urate is your fastest daily hydration read: smooth and creamy generally means a well-hydrated snake, and dry, hard or gritty is the signal to act. Dehydration is also on the list of causes of regurgitation, prolapse and egg binding, so it is worth getting right.

#### A SOAKING SNAKE

Soaking now and then, especially before a shed, is normal. A snake that suddenly will not leave its water dish is worth a close look at the eyes, chin and vent for mites.

## Setup checklist & targets

Tick each line before the snake goes in, and run through it again whenever something seems off.

### Enclosure

- ☐ Sized for the sex: at least 30×13×13 in (76×33×33 cm) for a male, 36×18×16 in (91×46×41 cm) for a female
- ☐ Front-opening, with a lid that latches or clamps. No tape anywhere
- ☐ Glass sides covered on three sides
- ☐ Out of the kitchen and out of a sunny window

### Heat and light

- ☐ Low-wattage halogen over the screen at the warm end, flat stone beneath
- ☐ Every heat source on a thermostat matched to it; probe placed per its instructions
- ☐ Lamp at least 4 to 6 in (10 to 15 cm) above the highest reachable spot
- ☐ Two digital thermometers, plus an infrared thermometer
- ☐ Lights on a 12-hour timer. No light at night
- ☐ Optional UVB: T5 HO 5.0 or 6%, about two-thirds of the enclosure length, over screen, not glass

### Inside

- ☐ 3 to 6 in (8 to 15 cm) of dry soil and sand mix, deepest at the cool end
- ☐ Warm hide, cool hide, and a humid hide
- ☐ Cork, branches and leaf litter for cover
- ☐ A water dish large enough to get into
- ☐ Digital hygrometer reading 30 to 50%

### Feeding and records

- ☐ Separate lidded feeding tub, lined with paper towel
- ☐ Feeding tongs and a snake hook
- ☐ Frozen mice sized to the snake, in a freezer away from human food
- ☐ Kitchen scale in grams, and the owner log

### Before it comes home

- ☐ State and local rules checked, pages 22 to 24
- ☐ Reptile vet found, first exam booked within 72 hours
- ☐ Quarantine room ready if you keep other reptiles
- ☐ Everything running and measured for a day

### Targets

| READING         | TARGET                                                                                        |
|-----------------|-----------------------------------------------------------------------------------------------|
| Basking surface | 90 to 95°F (32 to 35°C)                                                                       |
| Cool side       | 70 to 75°F (21 to 24°C)                                                                       |
| Night           | Room temperature. Add heat with no light below 60°F (16°C)                                    |
| Humidity        | 30 to 50%, moisture only in the humid hide                                                    |
| Light           | 12 hours on, 12 off                                                                           |
| Feeding         | Every 5 to 7 days young; about every two weeks as an adult; 10 to 14 days for a mature female |

## Emergency & quick targets

### CALL THE VET NOW IF

Open-mouth breathing, wheezing, drooling or mucus. No droppings with bloating and lethargy. Belly scales discolored or blistered and spreading. Red spots, saliva or cheesy material in the mouth. Mites. A burn. Anything protruding from the vent: keep it moist, do not push it back. A meal brought back up twice. Weight down more than 10%. Limp or unresponsive at normal temperature. A female past her laying window who is weak.

| READING                                             | °F                    | °C               |
|-----------------------------------------------------|-----------------------|------------------|
| Basking surface                                     | 90 to 95              | 32 to 35         |
| Cool side                                           | 70 to 75              | 21 to 24         |
| Night                                               | Room temperature      | Room temperature |
| <b>Add heat if the night falls below</b>            | <b>60</b>             | <b>16</b>        |
| <b>A fault if the basking surface is well above</b> | <b>95</b>             | <b>35</b>        |
| Humidity                                            | 30 to 50%. No misting |                  |

### WHILE YOU WAIT FOR THE VET

Keep the snake at its normal temperatures, toward the warm end of the range for a breathing problem. Isolate it from any other reptile on paper towel, with its own tools. No food, no handling, no home medication of any kind. A bite that breaks your skin: cool water over the snake's face to release it, then a doctor. Page 16.

### Your numbers

Reptile vet \_\_\_\_\_

Normal weight (g) \_\_\_\_\_

24-hour emergency clinic that sees reptiles \_\_\_\_\_

Last meal \_\_\_\_\_

Snake's name, sex and age \_\_\_\_\_

Last vet visit \_\_\_\_\_

### Routine, written down

Feeding day and prey \_\_\_\_\_

Lights on and off at \_\_\_\_\_

Bulb or UVB tube installed \_\_\_\_\_

Last full substrate change \_\_\_\_\_

### HAVE THIS READY WHEN YOU CALL

What you saw and when it started. All three temperatures and the humidity. The last meal and whether it stayed down. The last dropping. The weight record. Any new reptile in the house. Bring a fresh stool sample in a sealed bag, photos of the setup, and the owner log.

## Budget: setup & shopping list

United States retail ranges, last checked October 2026 and rounded to the nearest \$5. The low end is basic but correct gear; the high end is premium.

| INITIAL SETUP                                                                                                                                   | TYPICAL COST            |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Enclosure, 36×18×18 in (91×46×46 cm) or 2×2×2 ft, front-opening and latching; check the dimensions before you buy                               | \$100 to \$260          |
| Low-wattage halogen basking bulb, 50 W                                                                                                          | \$15 to \$25            |
| Dome fixture for the basking bulb                                                                                                               | \$10 to \$20            |
| Dimming thermostat                                                                                                                              | \$20 to \$35            |
| Plug-in timer for the lights, 12 hours on                                                                                                       | \$5 to \$15             |
| Digital thermometer and hygrometer                                                                                                              | \$15 to \$25            |
| Infrared thermometer                                                                                                                            | \$15 to \$30            |
| Soil-based substrate, aspen or coconut fiber; deep substrate can take more than one bag                                                         | \$15 to \$35            |
| Reptile-safe sand, for the 30% of the mix                                                                                                       | \$5 to \$15             |
| Warm, cool and humid hides                                                                                                                      | \$15 to \$85            |
| Branches for cover                                                                                                                              | \$10 to \$30            |
| Leaf litter for surface cover                                                                                                                   | \$10 to \$20            |
| Water dish the snake can get into                                                                                                               | \$5 to \$30             |
| Lidded plastic feeding tub                                                                                                                      | About \$10              |
| Feeding tongs                                                                                                                                   | \$5 to \$15             |
| Snake hook                                                                                                                                      | \$5 to \$15             |
| Kitchen scale in grams                                                                                                                          | \$10 to \$20            |
| First pack of frozen-thawed mice, sized to the snake                                                                                            | \$10 to \$25            |
| Optional: low-output T5 HO UVB fixture about two-thirds the enclosure length (check the length against your enclosure), 5.0 or 6% tube included | \$85 to \$100           |
| Optional: night heat if the room drops below 60°F (16°C), a heat mat and on/off thermostat; check the mat size against your enclosure           | \$25 to \$40            |
| <b>Equipment, every row added up</b>                                                                                                            | <b>\$390 to \$850</b>   |
| First wellness exam (\$50 to \$100) and fecal test (\$25 to \$50)                                                                               | \$75 to \$150           |
| <b>All in with the first vet visit, before the snake</b>                                                                                        | <b>\$465 to \$1,000</b> |

**Highly recommended, not in the total:** a UV index meter, about \$240 to \$265, the only way to check the UVB tube's height and when it needs replacing; UV cards do not measure UVB.

Without the optional UVB and night heat, the equipment is \$280 to \$710. A UVB tube's wattage is not its length: a slim high-output tube of 24 watts is about 22 in (56 cm) long, 39 watts about 34 in (86 cm), 54 watts about 46 in (117 cm). Match the length in inches to two-thirds of the enclosure. The snake is extra: \$50 to \$100 for a normal, wild-type, captive-bred animal, which brings the all-in figure to \$505 to \$1,080. A female costs \$50 to \$200 more than a male, and a rare color form far more. A headline figure of \$200 to \$500 for a hognose setup is common; the list above is the same kind of gear, priced line by line, with every tool this package uses.

# Budget: yearly costs

What a hognose costs to keep each year once the setup on the previous page is in place, on the same rounding.

| YEARLY ONGOING                                                      | TYPICAL COST          |
|---------------------------------------------------------------------|-----------------------|
| Frozen-thawed mice                                                  | \$60 to \$120         |
| Substrate replacement                                               | \$30 to \$50          |
| Electricity for heat                                                | \$40 to \$70          |
| UVB tube replacement, once a year if you run one                    | \$20 to \$35          |
| Yearly wellness exam                                                | \$50 to \$90          |
| <b>Yearly total, every row added up, about \$15 to \$35 a month</b> | <b>\$200 to \$365</b> |

Prices move; treat these tables as a snapshot, not a quote.

## PLAN FOR, DON'T COUNT ON AVOIDING

A radiograph is \$100 to \$250 a view, and treating a respiratory infection or an impaction can run into the hundreds. A few hundred dollars set aside matters as much as any line here.

## OVER A WHOLE LIFE

At \$200 to \$365 a year over the usual 10 to 15 years, a hognose costs roughly \$2,000 to \$5,475 in routine care after setup, before a single emergency. Cheap to buy and cheap to run, and a long time to keep.

## First 30 days

Most first-month problems come from doing too much: handling too soon, feeding too often, or fixing a hood and hiss that needs no fixing.

### Before it arrives

- ☐ Enclosure set up and running, heat, light and thermostat checked with both thermometers for a full day
- ☐ Humidity confirmed 30 to 50%
- ☐ Legal check done, and the receipt names *Heterodon nasicus*, captive-bred
- ☐ Vet chosen; first exam booked within 72 hours
- ☐ Asked the seller: hatch date, last meal, prey size, plain or scented

### Week 1

- ☐ Snake straight into its enclosure, or into quarantine if you keep other reptiles
- ☐ No handling. Two weeks to settle
- ☐ First vet exam with a fresh stool sample; record the baseline weight
- ☐ Read both thermometers and the hygrometer daily
- ☐ Fresh water daily; spot-clean droppings

### Week 2

- ☐ First feeding, in the feeding tub, on the seller's schedule and prey size
- ☐ Leave it alone for 48 hours after
- ☐ Refused? No panic. Page 13

### Week 3

- ☐ First handling session: lift out with a hook, from the side, no more than 5 minutes, end on a calm snake
- ☐ Not within 48 hours of a meal, not during a shed
- ☐ First weekly weigh-in in the log
- ☐ Probe check: still where it belongs?

### Week 4

- ☐ Second weigh-in. Is the line flat, rising or falling?
- ☐ Cross-section body condition check
- ☐ Review the feeding record against page 11
- ☐ Look over the droppings and urates, page 30

### Throughout

- ☐ Wash hands with soap after every contact; nothing cleaned in the kitchen sink
- ☐ No misting, ever
- ☐ Hooding, hissing, playing dead: step back and let it settle
- ☐ Quarantine running its full 3 to 6 months, if you keep other reptiles
- ☐ Sitter sheet, page 39, filled in before your first trip

#### THE ONE THING TO DO EVERY SINGLE DAY

Read the thermometers and the hygrometer and write them down. A thermostat that has failed, a probe that has slipped, or a dish that has spilled into the substrate all show up in those numbers before they show up in the snake.

## Symptom quick reference

Find what you are seeing, then turn to the page. When in doubt, call the vet.

| YOU SEE                                                     | LIKELY CAUSE                                                   | PAGE | HOW SOON                        |
|-------------------------------------------------------------|----------------------------------------------------------------|------|---------------------------------|
| Open-mouth breathing, wheezing, mucus                       | Respiratory infection                                          | 26   | Vet promptly                    |
| Discolored or blistered belly scales                        | Scale rot                                                      | 26   | Vet if past mild                |
| No droppings, bloating, lethargy                            | Impaction                                                      | 27   | Vet, not more waiting           |
| Meal brought back up                                        | Handling, cool temperatures, stress, oversized prey, parasites | 27   | Rest two weeks; vet if repeated |
| Skin folds, doughy feel, weight climbing                    | Obesity                                                        | 27   | Adjust feeding                  |
| Specks around the eyes and chin, long soaking               | Mites                                                          | 28   | Vet this week                   |
| Weight loss while eating, watery droppings                  | Internal parasites                                             | 28   | Vet with a sample               |
| Repeated regurgitation, weight loss, firm mid-body swelling | <i>Cryptosporidium</i>                                         | 28   | Vet, isolate now                |
| Shed in pieces, eye cap left on                             | Stuck shed                                                     | 29   | Vet for an eye cap              |
| Red spots, saliva or cheesy material in the mouth           | Mouth rot                                                      | 29   | Vet promptly                    |
| Damaged skin where it lay on the heat                       | Burn                                                           | 29   | Vet                             |
| Tissue outside the vent                                     | Prolapse                                                       | 29   | Emergency                       |
| Off food, buried, weight steady, late fall                  | Seasonal slowdown                                              | 21   | Weigh every 1 to 2 weeks        |
| Yellow, orange or gritty urates                             | Dehydration, then kidney trouble                               | 30   | Water first; vet if it persists |
| Blood in droppings, or straining                            | Bleeding, blockage, cloaca problem                             | 30   | Vet the same day                |
| Female past laying, weak, listless                          | Egg binding                                                    | 20   | Vet now                         |
| Hood, hiss, rolling over to play dead                       | Normal defense                                                 | 15   | Step back                       |
| Limp, unresponsive, no trigger                              | Not the act. Illness or temperature                            | 15   | Vet the same day                |
| Refusing food, no other sign                                | Shed, season, breeding, setup, picky                           | 13   | Weigh and work the list         |

This table points to a page; it does not diagnose. Several different problems share the same early signs, which is what the vet visit is for.

## Daily, weekly & seasonal routine

A low-maintenance snake, once the routine is written down.

### Daily

- ☐ Read both thermometers and the hygrometer
- ☐ Read the second thermometer away from the thermostat probe
- ☐ Empty, wash and refill the water dish
- ☐ Pull droppings and urates as you see them; note what they looked like
- ☐ Check lights came on and went off on time
- ☐ Wash hands with soap after any contact

### Weekly

- ☐ Weigh in grams and write it down
- ☐ Cross-section body condition check
- ☐ Check the thermostat probe has not moved
- ☐ Spot-check surfaces with the infrared thermometer
- ☐ One or two short handling sessions, never within 48 hours of a meal

### Every feeding

- ☐ Thaw in the refrigerator; never the microwave or the kitchen sink
- ☐ Feed in the tub, from tongs; record the date, prey and result
- ☐ Wash hands, tongs and tub

### Every shed

- ☐ Moisten the humid hide substrate when the eyes cloud
- ☐ Leave it alone until the shed is off
- ☐ Check the skin for both eye caps and the tail tip

### Every three months

- ☐ Replace all substrate and disinfect, if the bed stayed dry and clean. Monthly if it went damp or smelly
- ☐ Check the frozen mouse stock date

### Yearly

- ☐ Wellness exam and fecal test
- ☐ Replace the UVB tube, if you use one, on the maker's schedule
- ☐ Review prey size against current weight

### Seasonal

- ☐ **Fall:** expect a slowdown; weigh every 1 to 2 weeks while it lasts
- ☐ **Winter:** watch night temperatures as the house cools; add heat with no light below 60°F (16°C)
- ☐ **Spring:** appetite should return. Still refusing by spring is a vet visit
- ☐ **Summer:** a hot room pushes the basking surface up; check it is not well above 95°F (35°C)

## Power outages, travel & transport

Heat is the one thing a snake cannot make for itself. For a hognose the line to hold is 60°F (16°C).

A healthy adult snake tolerates one ordinary cool night without harm. The risk is duration: sustained lower temperatures leave a reptile more open to infection and can stop it eating. For a hognose, room temperature is a normal night, and 60°F (16°C) is the point to add heat, move the snake, or call the sitter.

### When the power goes out: do

- Put a battery thermometer in the enclosure so you know the real number.
- Drape thick towels or a blanket over it, leaving a vent gap.
- Or move the snake in a ventilated tub to the warmest room, usually one without outside walls.
- A wrapped hot water bottle or hand warmer beside the snake, **never touching it**.
- Stop feeding until heat is reliably back. Keep water available.
- More than a day or two below 60°F (16°C): board it with a vet or a friend with power.

### Don't

- Use candles, a gas stove or any open flame near the enclosure: a fire and carbon monoxide risk.
- Put a heat pack against the snake. A cold, sluggish snake cannot move away in time.
- Feed to "keep its strength up." Food sitting in a cold gut is a risk.
- Warm a cold, unresponsive snake in hot water.

### Call the vet if

Open-mouth breathing, wheezing or clicking, mucus at the nose or mouth, or a snake that stays limp once it is back at normal temperature.

## Leaving town

| TRIP          | ADULT                                                                                                                                                | JUVENILE, OR A SNAKE RECOVERING FROM ILLNESS          |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 1 to 2 nights | Often fine alone: lights on timers, thermostat running, fresh water                                                                                  | A sitter once a day for water and a temperature check |
| 3 to 7 nights | A sitter every 1 to 2 days for water, droppings and a temperature check. Feed on the normal schedule                                                 | A sitter daily, with written instructions             |
| Over a week   | A daily sitter who has done a practice visit with you, or boarding with a reptile vet or reptile shop. Never a general dog and cat boarding facility |                                                       |

An adult fed every two weeks usually needs no feeding at all on a trip of a week. Time the meal before you leave, then let the sitter skip it.

## Transport and vet trips

- A ventilated plastic tub with a towel to grip, inside a dark bag or box. No loose substrate and no water dish.
- Skip the meal before a trip. Warm the car first in cold weather, and on a trip over about 30 minutes below 65°F (18°C), add a wrapped hand warmer beside the tub.
- Check it every 20 to 30 minutes at a stop. Never leave a snake in a parked car, hot or cold, even for a few minutes.

### OUTAGE KIT, PACKED BEFORE YOU NEED IT

Two or three hand warmers or a hot water bottle, two thick towels, a battery thermometer, a ventilated tub, and the vet and emergency clinic numbers on paper, not only in a phone that might die.

## FILL IN AND LEAVE BY THE ENCLOSURE

# Pet-sitter sheet

Everything a sitter needs on one page. Walk them through it in person once before the first solo visit.

Snake's name \_\_\_\_\_

Normal weight and look \_\_\_\_\_

My phone \_\_\_\_\_

Reptile vet \_\_\_\_\_

24-hour emergency clinic \_\_\_\_\_

Backup contact \_\_\_\_\_

### Each visit

- ☐ Empty, wash and refill the water dish
- ☐ Read both thermometers and the hygrometer, and write them down
- ☐ Check the lights are on in the day and off at night
- ☐ Remove any droppings with the scoop; note what they looked like
- ☐ Wash your hands with soap after, away from the kitchen
- ☐ It may be buried and out of sight. That is normal

### Feeding while I'm away

Feed on (date) \_\_\_\_\_

Prey and size \_\_\_\_\_

Where the mice and tub are \_\_\_\_\_

Or: no feeding this trip \_\_\_\_\_

### Normal readings

|                 |                                           |
|-----------------|-------------------------------------------|
| Basking surface | 90 to 95°F (32 to 35°C)                   |
| Cool side       | 70 to 75°F (21 to 24°C)                   |
| Night           | Room temperature, never below 60°F (16°C) |
| Humidity        | 30 to 50%. Do not mist                    |

### Please don't

- Pick the snake up unless we have discussed it. If it hoods, hisses or rolls over and plays dead, that is a normal bluff: step back and leave it.
- Change any dial, timer or thermostat, mist the enclosure, or feed anything not on this sheet.
- Handle frozen mice near human food, or thaw them in the microwave or kitchen sink.

#### CALL ME IF

The night reading is under 60°F (16°C) or the basking surface well over 95°F (35°C). The lights are off in the day. Open-mouth breathing, wheezing or mucus. Something sticking out of the vent. It is limp at normal temperature, or has not moved across two visits while out in the open. It has escaped. If I don't answer, call the vet number above.

## Owner log

[illegible]

## WHY BOTHER LOGGING

A hognose that is getting sick often shows it on the scale before it shows it anywhere else. Sixteen rows is four months, long enough to see a real trend rather than a bad week, and it is the first thing a vet will want to see. For an adult, it is also how you notice the slow climb toward obesity.

# Equipment, quarantine & vet log

## Equipment log

| ITEM                                              | INSTALLED OR OPENED | REPLACE OR CHECK BY | DONE |
|---------------------------------------------------|---------------------|---------------------|------|
| Halogen basking bulb                              |                     |                     |      |
| Thermostat (check against a second thermometer)   |                     |                     |      |
| Thermometer and hygrometer batteries              |                     |                     |      |
| UVB tube, if used (6 to 12 months, per the maker) |                     |                     |      |
| Full substrate change (about every 3 months)      |                     |                     |      |
| Frozen mouse stock (use within the year)          |                     |                     |      |

## Twelve-month planner

M1 to M12 are months 1 to 12, starting with whichever month you're in. Tick the box in the month you did it.

| TASK                    | M1                       | M2                       | M3                       | M4                       | M5                       | M6                       | M7                       | M8                       | M9                       | M10                      | M11                      | M12                      |
|-------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Body condition check    | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Prey size reviewed      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Full substrate change   | <input type="checkbox"/> |                          |                          | <input type="checkbox"/> |                          |                          | <input type="checkbox"/> |                          |                          | <input type="checkbox"/> |                          |                          |
| UVB tube, if used       | <input type="checkbox"/> |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          |
| Vet exam and fecal test | <input type="checkbox"/> |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          |                          |

## Quarantine record

Arrived \_\_\_\_\_

Fecal test results \_\_\_\_\_

From \_\_\_\_\_

Clock restarted on \_\_\_\_\_

Species on the receipt \_\_\_\_\_

Cleared on (3 to 6 months) \_\_\_\_\_

## Vet visit log

| DATE | REASON | FINDINGS AND TESTS | TREATMENT, FOLLOW-UP |
|------|--------|--------------------|----------------------|
|      |        |                    |                      |
|      |        |                    |                      |
|      |        |                    |                      |
|      |        |                    |                      |

# Glossary, A to D

|                                   |                                                                                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Albino, leucistic, xanthic</b> | Bred color forms: albino lacks dark pigment, leucistic is pale overall, xanthic is yellow. Indiana exempts captive-bred animals in these forms. Page 23. |
| <b>Ambient humidity</b>           | The humidity of the enclosure air in general, as opposed to inside the humid hide. 30 to 50% for this species. Page 9.                                   |
| <b>Anemia</b>                     | Too few red blood cells, here from blood lost to mites. Page 28.                                                                                         |
| <b>Antiparasitic</b>              | A drug that kills parasites. The vet chooses it and sets the dose.                                                                                       |
| <b>Basking surface</b>            | The flat spot under the heat lamp where the snake warms up: 90 to 95°F (32 to 35°C).                                                                     |
| <b>Bioactive</b>                  | A planted substrate with a cleanup crew of tiny invertebrates that break down waste.                                                                     |
| <b>Brumation</b>                  | The reptile version of a winter slowdown: less activity, less appetite, slower metabolism. Not needed by a pet. Page 21.                                 |
| <b>Captive-bred</b>               | Hatched from parents kept by people, rather than caught in the wild. Several state laws turn on it.                                                      |
| <b>Cellulitis</b>                 | An inflamed infection of the skin and the tissue under it. A sign in the 2009 bite case. Page 16.                                                        |
| <b>Ceramic heat emitter</b>       | A screw-in heater that gives off heat with no visible light, usable at night.                                                                            |
| <b>Cloaca and vent</b>            | The cloaca is the single chamber that waste, urates and eggs all pass through; the vent is its opening under the tail.                                   |
| <b>Color form</b>                 | A line bred for a color or pattern, also called a morph. The care is identical; only the price changes.                                                  |
| <b>Colubrid</b>                   | A member of the snake family Colubridae, the family the hognose is filed under in New York City's rule. Page 22.                                         |
| <b>Conditional</b>                | A legal status: allowed without a permit only if a stated condition is met. Page 23.                                                                     |
| <b><i>Cryptosporidium</i></b>     | A single-celled stomach parasite of snakes with no reliable cure. Page 28.                                                                               |
| <b>D3</b>                         | Vitamin D3, the form of vitamin D a reptile's skin makes when UVB reaches it. Page 9.                                                                    |
| <b>Death feigning</b>             | Playing dead: rolling belly up, mouth open, tongue out. Also called thanatosis. Normal. Page 15.                                                         |
| <b>Devenomized</b>                | Venom glands removed or disabled. New York City bans a hognose even if it has been. Page 22.                                                             |
| <b>Dimming thermostat</b>         | A thermostat that turns a heat lamp's power up and down like a light dimmer. The match for a halogen. Page 8.                                            |
| <b>Duvernoy's gland</b>           | The gland in the upper jaw that produces a hognose's mild venom, feeding the rear fangs. Page 16.                                                        |
| <b>Dysecdysis</b>                 | A bad or stuck shed. Page 29.                                                                                                                            |
| <b>Dystocia</b>                   | Egg binding: a female unable to pass her eggs. An emergency. Page 20.                                                                                    |

# Glossary, E to N

|                             |                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Ecdysis</b>              | Shedding the skin. Page 29.                                                                                                |
| <b>Enrichment</b>           | Features that let an animal do what it naturally does: for a hognose, above all, digging. Page 17.                         |
| <b>Envenoming</b>           | Venom actually entering the body in a bite. Page 16.                                                                       |
| <b>Eye cap</b>              | The clear scale over each eye, also called the spectacle. It sheds with the skin; one left on needs a vet.                 |
| <b>Fecal test</b>           | A lab check of a stool sample for parasites. Part of every new snake's first vet visit.                                    |
| <b>Ferguson zone</b>        | A ranking of reptiles by how much ultraviolet light they seek in the wild, zone 1 to zone 4. Hognoses are zone 2. Page 9.  |
| <b>Frozen-thawed</b>        | Prey bought frozen and thawed before feeding. The standard for this species. Page 12.                                      |
| <b>Fuzzy, peach fuzzy</b>   | A fuzzy is a young mouse with its first fur; a peach fuzzy sits between a pinky and a fuzzy. Page 11.                      |
| <b>Gout</b>                 | A painful disease from uric acid building up in the joints or organs. High-protein feeding and dehydration push toward it. |
| <b>Gradient</b>             | The spread of temperatures from the warm end to the cool end, so the snake can choose. Page 7.                             |
| <b>Gravid</b>               | Carrying eggs. Page 20.                                                                                                    |
| <b>Halogen basking bulb</b> | A bright, focused bulb that heats the spot beneath it. The usual day heat for this species.                                |
| <b>Hemipenes</b>            | A male snake's paired reproductive organs, kept inside the base of the tail. What probing and popping detect.              |
| <b>Hide</b>                 | A box, cave or piece of cork bark the snake can get fully inside, out of sight. Page 6.                                    |
| <b>Hooding</b>              | Flattening the neck into a cobra-like hood as a bluff. Page 15.                                                            |
| <b>Hopper</b>               | A young, weaned mouse, bigger than a fuzzy.                                                                                |
| <b>Humid hide</b>           | A hide with moistened substrate or moss inside, the one damp spot in a dry enclosure. Page 9.                              |
| <b>Hygrometer</b>           | A humidity meter. Digital, read at snake level.                                                                            |
| <b>Impaction</b>            | A gut blocked by swallowed substrate or other material. Page 27.                                                           |
| <b>Incubation</b>           | Keeping eggs warm until they hatch. Page 20.                                                                               |
| <b>Infrared thermometer</b> | A handheld point-and-read thermometer for surface temperatures.                                                            |
| <b>Lymph nodes</b>          | Small glands that help fight infection; they swelled in the 2009 bite case.                                                |
| <b>Mites</b>                | Tiny blood-feeding parasites, <i>Ophionyssus natricis</i> in snakes. Page 28.                                              |
| <b>Musk</b>                 | A foul-smelling fluid released from the vent in defense.                                                                   |
| <b>Native species</b>       | A species that lives wild in that state. Native-wildlife law is what restricts hognoses most often.                        |
| <b>Nongame species</b>      | Wildlife that is not hunted for sport or food. Several states cap or protect native nongame reptiles.                      |

# Glossary, O to Z

|                                                                  |                                                                                                                                         |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ordinance</b>                                                 | A city or county law. It can be stricter than the state.                                                                                |
| <b>Permethrin</b>                                                | An insecticide. Some permethrin products are highly toxic to hognoses: mite treatment is the vet's call.                                |
| <b>Pinky</b>                                                     | A newborn, hairless mouse, the first food for a hatchling.                                                                              |
| <b>Probing and popping</b>                                       | The two hands-on ways to sex a snake. Vet or experienced breeder only. Page 19.                                                         |
| <b>Prolapse</b>                                                  | Tissue pushed out through the vent. An emergency. Page 29.                                                                              |
| <b>Pulse proportional thermostat</b>                             | A thermostat that sends short pulses of power. The match for a ceramic heat emitter.                                                    |
| <b>PVC</b>                                                       | Polyvinyl chloride, the rigid plastic many enclosures are made of.                                                                      |
| <b>Quarantine</b>                                                | Keeping a new animal apart from other reptiles for 3 to 6 months. Page 18.                                                              |
| <b>Radiograph</b>                                                | An X-ray image.                                                                                                                         |
| <b>Rear-fanged</b>                                               | With enlarged fangs toward the back of the upper jaw rather than at the front.                                                          |
| <b>Regurgitation</b>                                             | A swallowed meal brought back up. Page 27.                                                                                              |
| <b>Rostral scale</b>                                             | The scale at the tip of the snout, upturned and shovel-shaped in a hognose for digging.                                                 |
| <b><i>Salmonella</i></b>                                         | Bacteria healthy reptiles carry and shed in droppings. Hands and surfaces to mouth is the risk. Page 14.                                |
| <b>Scale rot</b>                                                 | A belly-skin infection from damp, dirty substrate; blister disease is its early stage, with pustules (small pus-filled bumps). Page 26. |
| <b>Scenting</b>                                                  | Rubbing prey with a scent such as fish or toad to tempt a refusing snake. Page 13.                                                      |
| <b>Species of special concern, of greatest conservation need</b> | State conservation listings below threatened. Several states bar keeping the eastern hognose because of them.                           |
| <b>Stomatitis</b>                                                | Mouth rot: infection of the tissue lining the mouth. Page 29.                                                                           |
| <b>Substrate</b>                                                 | The loose material on the enclosure floor. Page 10.                                                                                     |
| <b>T5 HO</b>                                                     | A slim fluorescent tube 5/8 in across (T5) in the high-output (HO) format. The usual UVB source.                                        |
| <b>Thermostat</b>                                                | A controller that cuts the heat at a set temperature. Non-negotiable. Page 8.                                                           |
| <b>Ultrasound</b>                                                | Imaging with sound waves. A non-invasive way for a vet to sex a snake or find eggs.                                                     |
| <b>Units</b>                                                     | in, inches; ft, feet; cm, centimeters; mm, millimeters; g, grams; °F and °C, degrees Fahrenheit and Celsius; %, percent.                |
| <b>Urates</b>                                                    | The white, chalky paste of uric acid passed with a reptile's droppings. Page 30.                                                        |
| <b>UVB</b>                                                       | Ultraviolet B, the part of sunlight a reptile's skin uses to make vitamin D3. Page 9.                                                   |
| <b>UVI</b>                                                       | Ultraviolet index, a measure of UVB strength at one spot. 1.1 to 3.0 where a hognose basks.                                             |
| <b>Wild-type</b>                                                 | The natural color and pattern, not a bred color form. The cheapest hognose to buy.                                                      |

## Sources

Where the numbers in this package come from, so you can check them yourself.

### Beastly Facts guides drawn on

The hognose snake enclosure setup, feeding, health issues, handling, enrichment, cost and legal guides, and the hognose encyclopedia entry, carry every core husbandry figure: enclosure sizes by sex and age, temperatures, humidity, substrate, prey sizes and schedules, the refusal plan, the regurgitation plan, the display, the venom summary, the handling rules, setup and running costs and lifespan. The shared reptile guides carry the rest: quarantine and choosing a healthy reptile, heating and thermostats, the emergency plan with its hognose temperature row, snake sexing and body condition, snake brumation, shedding, *Salmonella* and hygiene with the cleaning schedule, reading stool and urates, and UVB lighting. The ball python feeding and health guides supply the thawing and storage method and the prolapse first aid, which do not depend on the species. The legal section comes from the Beastly Facts state legal research, checked 1 October 2026.

### Research on this species

- Weinstein and Keyler (2009), *Toxicon* 54(3): Local envenoming by the Western hognose snake (*Heterodon nasicus*): a case report and review of medically significant *Heterodon* bites. The bite case, its five-month recovery, and the conclusion that hognoses should not be considered dangerous.
- Damm, Vilcinskas, Kreuels and Lüddecke (2026), *Frontiers in Amphibian and Reptile Science*: 34 reports of snakebite envenoming by the Western Hognose snake, a retrospective online survey. Bite counts, feeding context, hold time against symptoms, recovery times and the keeper recommendations.
- Durso and Mullin, *Ethology* 120: Intrinsic and extrinsic factors influence expression of defensive behavior in Plains Hog-nosed Snakes. Death feigning by size, sex and temperature.
- Nagabaskaran, Skinner and Miller (2022), *Animals* 12(23): Western hognose snakes prefer environmental enrichment. The preference study.
- Kerns, Animal Diversity Web, University of Michigan: *Heterodon nasicus*. Range, habitat, length and weight, wild lifespan, toads in the diet, clutch size and hatching time, the winter season, and the state listings.
- Galewood, Reptiles Magazine: Western Hognose Snake Care and Breeding. Adult lengths and weights by sex, hatchling weights, the pre-lay shed and laying box, incubation, and the warnings on permethrin and *Cryptosporidium*.
- Healey, ReptiFiles: the hognose snake care guide and its pages on temperatures, humidity and lighting, handling and body language, brumation, and mites. The middle temperature band and 60°F (16°C) night floor, the basking stone, the UVB specification, cool water for a bite that holds on, and the brumation schedule.

### Veterinary references

- Merck Veterinary Manual, pet owner version: Disorders and Diseases of Reptiles. Respiratory infection, mouth rot, mites and the enclosure clean, blister disease, gout, prolapse, and egg binding diagnosis and treatment.
- Merck Veterinary Manual: Parasitic Diseases of Reptiles. *Cryptosporidium* signs, diagnosis and treatment, the permethrin licensed for reptiles, and the role of quarantine.
- Divers (1996), Proceedings of the Association of Reptilian and Amphibian Veterinarians: constipation in snakes. The missed stool that comes before a snake stops eating.
- Zoo Med Laboratories: the ReptiSun 5.0 and 10.0 UVB T5 HO product pages. Tube lengths and wattages.
- Spyder Robotics, Herpstat thermostat manual: probe placement, and the warning against aluminum tape on the probe.
- VCA Animal Hospitals (originally Veterinary Centers of America): Common Snake Health Issues. Egg binding causes and signs, retained eye caps, burns and their causes, and the causes of a lost appetite.

The Beastly Facts guides cite further veterinary sources of their own, among them the Merck Veterinary Manual's reptile husbandry, nutrition and quarantine pages, VCA's snake housing page, the Centers for Disease Control and Prevention on reptiles, amphibians and frozen feeder rodents, and Iowa State University's biosecurity guidance.

## REFERENCE

# Where the sources disagree

Where credible sources give different numbers, this package prints one and says why.

| TOPIC                | WHAT THE SOURCES SAY                                                                                                                                                                                                    | WHAT THIS PACKAGE USES                                                                    |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Night temperature    | Room temperature, no colder than 60°F (16°C), with no night heat needed (a specialist care sheet and the setup guide); a snake needs no night heat if the room holds 65 to 70°F (18 to 21°C) (the shared heating guide) | Room temperature, adding heat with no light below 60°F (16°C)                             |
| Adult length         | Males 16 to 24 in, females 21 to 30 in (a breeder); 14 to 37 in, about 20 in average (a university account); 14 to 46 in for the genus; 1.5 to 3.5 ft for all hognoses; 2 to 3.5 ft                                     | The breeder's split by sex, because enclosure size depends on it                          |
| Lifespan             | 10 to 15 years commonly, 15 to 20 reported in captivity; 10 to 15 with one known 18-year-old; about 14 in the wild                                                                                                      | 10 to 15 years, 15 to 20 reported                                                         |
| Activity             | Active by day, most of all morning and late afternoon; or most active at dawn and dusk                                                                                                                                  | Active by day, so a 12-hour light cycle                                                   |
| Quarantine           | 3 to 6 months (veterinary); 60 to 90 days (hobby floor); 2 to 3 months (a care sheet)                                                                                                                                   | 3 to 6 months                                                                             |
| Mite treatment       | A care sheet describes a home treatment with a diluted human lice product; a breeder warns permethrin products are highly toxic to hognoses; veterinary medicine uses a permethrin licensed for reptiles                | Vet-only treatment, never a store product on your own                                     |
| Weighing             | Weekly (the hognose health guide and a care sheet); monthly for a growing juvenile (the shared body condition guide)                                                                                                    | Weekly                                                                                    |
| Heat source          | A halogen over the substrate on a dimming thermostat (the setup and cost guides)                                                                                                                                        | Halogen by day; a mat or emitter only if the room drops below 60°F (16°C) at night        |
| A bite that holds on | Remove the snake quickly (a bite survey); do not pull it off, pour cool water on its face (the handling guide, a care sheet)                                                                                            | Both: quickly, with cool water, never by yanking                                          |
| UVB tube life        | Every 12 months (a care sheet); 6 to 12 months by model (the UVB guide)                                                                                                                                                 | The maker's schedule, no longer than 12 months                                            |
| Costs                | Setup \$200 to \$500 (a common headline figure); priced line by line with every tool this package uses, \$390 to \$850, and \$200 to \$365 a year (the cost guide)                                                      | Itemized: \$390 to \$850 of equipment, \$200 to \$365 a year (about \$15 to \$35 a month) |

## Version history & about

What changed in each edition, so a printed copy can be placed.

### Version history

This is edition 1.1 (Oct 2026). The store page shows the current edition; re-download free if yours is older. Where this list and a page disagree, the page wins.

- **1.1, Oct 2026.** Aligned to the current layout: cover icon, section order and names, page pointers, a separate page for where the sources disagree, and added glossary entries on a third glossary page. Costs: every item priced and rounded to the nearest \$5; setup \$390 to \$850, yearly \$200 to \$365, the budget now over two pages. Other changes are wording only.
- **1.0, Oct 2026.** First edition, 44 pages.

---

This package is a husbandry reference compiled from the listed sources and cross-checked against current veterinary guidance at the time of publication. It is not a substitute for a reptile veterinarian, and nothing in it should delay a vet visit. Treatments described here are for recognizing what a vet may do, never for use at home, and every dose is set by the vet. The law section is a summary as of October 2026, not legal advice.

© Beastly Facts · Edition 1.1 · October 2026 · For personal use by the purchaser. Please don't redistribute.